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Recruit Ease

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Abstract

In the fast paced modern industry, manual recruitment techniques are becoming inefficient, often bypassing qualified candidates and consuming significant time. Recruit Ease, our web based solution intends to use the capabilities of Large Language Models (LLMs) through prompt engineering, to automate key recruitment stages. In this way, the project would streamline the time-consuming tasks such as initial resume filtering and would also elevate the quality of the interview feedback and analysis along with applicant communication. This would redefine the way organizations approach talent acquisition, ensuring a future where the recruitment process is characterized by efficiency, fairness and unparalleled precision. By combining innovation with efficiency, Recruit Ease aims to become a competitive player in the growing market of Application Tracking Systems (ATSs). It will position itself as a promising solution to modernize and enhance the recruitment industry.

Executive Summary

In today's rapidly evolving industry, where time is a valuable asset and talent is in high demand, traditional recruitment processes have become lengthy, biased, and inefficient. In the presence of these limitations there is a need for effective recruitment methods utilizing cutting-edge technologies. Recruit Ease is a web-based solution enhancing the recruitment process by harnessing the capabilities of Large Language Models (LLMs) through effective prompt engineering. The application aims to enhance efficiency, fairness, and communication in talent acquisition by automating key recruitment stages like resume parsing, interview feedback and candidate interaction. While it aligns with the Sustainable Development Goal of "decent work and economic growth," challenges like LLM limitations and reluctance to change from traditional methods may arise.

To examine the potential, methodology, and challenges of Recruit Ease extensive literature review was conducted. Firstly, we analyzed existing applications of artificial intelligence in optimizing recruitment processes, both in terms of administrative efficiency and enhanced candidate experience. This was followed by the study of Natural Language Processing (NLP) and the Transformer model's role in LLMs. Further research was done on prompting techniques, including few-shot prompting, chain of thought prompting, and the ReAct methodology. Finally, the utilization of Langchain and vector databases in the construction of LLM applications was explored.

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Chapter 1 Introduction

In the world of recruitment, the age-old saying, “time is money”, holds profound relevance. The ability to attract and retain the right talent is of high importance to any organization. However, the traditional recruitment landscape, dominated by manual processes and personal judgments, has often lagged behind in both efficiency and effectiveness. The conventional approach of resume sifting and interview analysis is slow and has the potential to unintentionally overlook capable candidates. Moreover, Human Resource (HR) departments face challenges to make an efficient, smooth, and effective recruitment process for both themselves and the applicants.

Considering the challenges, we are introducing a web-based solution, Recruit Ease, that will give all applicants a fair chance while making the recruitment process smooth and time-effective for both the applicants and the recruiters. Designed with the modern HR professional in mind, this platform harnesses the unparalleled capabilities of Large Language Models (LLMs) combined with the innovation of prompt engineering. This will lead us to successfully automate and refine key stages of the recruitment process from the initial shifting of resumes to the providing feedback post-interviews. By doing so, Recruit Ease promises organizations a future where recruitment is no longer a game of chance but a precise, efficient, and equitable process.

This transformative approach addresses the pressing challenges faced by HR departments worldwide. It promises a world where every applicant is judged on their merit, every resume is given its due consideration, and where the communication between recruiters and candidates is seamless and effective. With Recruit Ease, we envision a recruitment ecosystem that is not only time-effective but also ensures that organizations truly acquire the talent they seek, and deserving candidates find their rightful place in the professional world.

In the upcoming chapters of this report, we will explore the product vision, conduct a literature review, and examine relevant works essential for Recruit Ease.

1.1 Purpose of this Document

The purpose of this document is well-defined. It is to present a detailed overview, analysis, and potential of Recruit Ease. This report will analyze various domains of our project by offering in-depth insight in each aspect that will be used. Central to our research is the question: Can we harness the capabilities of LLMs through prompt engineering to create a tool that fundamentally improves the recruitment process? Through this report, we will delve into the methodology, design, functionality, and prospective value of Recruit Ease, while also addressing potential challenges and future enhancements.

1.2 Intended Audience

The document's intended audience spans a variety of individuals. Firstly, it caters to educational institutions, specifically professors, researchers, and students who show an interest in exploring the real-world applications of LLMs and prompt engineering. Secondly, it aims to serve recruitment industry stakeholders, encompassing HR professionals, recruitment agencies, and organizations that are in pursuit of efficient talent acquisition tools and strategies. Lastly, technology enthusiasts will find the document particularly insightful, especially in understanding the application and broader implications of cutting-edge technology within the recruitment sector.

1.3 Definitions, Acronyms, and Abbreviations

ATS: Application Tracking System

CoT: Chain of Thought

CoT-SC: Chain of Thought Self-consistency

e-HRM: Electronic Human Resource Management

HR: Human Resource

HRIS: Human Resource Information System

LLM: Large Language Models

LSTM: Long Short-Term Memory

NLG: Natural Language Generation

NLI: Natural Language Inference

NLP: Natural Language Processing

NLU: Natural Language Understanding

OOD tasks: Out of Domain tasks

TELeR: Turn, Expression, Level of Details, Role

VDBMS: Vector Database Management Systems

1.4 Conclusion

This document is carefully structured for clear understanding, providing the following overview. Chapter 1 begins with an introduction to the project and explains the document's purpose, target audience, and terms used. Chapter 2 provides a detailed look into the product vision, covering topics like problem overview, goals and objectives, scope, constraints, business opportunities and stakeholders description/user characteristics. Chapter 3 then focuses on a literature review of related work in the area.

Chapter 2 Project Vision

In today's rapidly evolving industry, where time is a valuable commodity and talent is in high demand, the traditional methods of manual recruitment have started to show their limitations. Therefore, Recruit Ease is a web-based solution that intends to use advanced technologies including the use of LLMs and prompt engineering, to automate key recruitment stages, effectively expediting processes, enhancing fairness and ensuring unparalleled precision in the process of talent acquisition.

2.1 Problem Domain Overview

Recruit Ease, will be a cutting-edge web based solution that intends to use the capabilities of LLMs and prompt engineering, to bring a revolution in the recruitment industry. The platform will cater to both recruiters and interviewers, providing them with a platform to ease out the entire process for them. The user can login into the application, upload CVs in bulk, add job descriptions and criteria. The system will intelligently parse the resumes and perform initial screening ensuring that only the most relevant and capable candidates are shortlisted for the next steps. Moreover, the interviewers will be allowed to add interview feedback against a pre-defined criteria and give their final verdict on whether the applicant should be selected for the next steps or not. They can do this while seeing the applicant profile in a concise way, ensuring a good user experience. In addition to these functionalities, the system provides a seamless communication between the recruiters and the applicants making sure that the applicants get customized emails. This innovative approach aims to streamline different stages of the recruitment process while ensuring efficiency, fairness and precision, making Recruit Ease an indispensable tool for modern talent acquisition.

2.2 Problem Statement

The project highlights the problem of having a proper web based application to facilitate different stages of the recruitment process and ensure effective communication with the applicants.

2.3 Problem Elaboration

In the globalized and fast-paced world today, the demand for talent is ever-growing. While companies want to select the best candidates that fit their job description the best, candidates themselves want to end up with jobs that are the best for their skill set and capabilities. Yet, despite the advent of technology and advancements, the recruitment process still faces inefficiencies and inconsistencies.

Firstly, the process of sifting through hundreds of resumes or relying on simple keyword matches is ineffective and has some loopholes. Therefore, the traditional ways can unintentionally overlook the qualified candidates just because they lacked the correct keywords or simply because their resume did not look visually appealing to the recruiter. Furthermore, effectively communicating the interview feedback to the concerned people in the recruitment cycle is a difficult and time taking process and thus many interviewers avoid communicating the complete feedback in order to save their time and just give their final verdict. Lastly, many HRs struggle to communicate with applicants effectively. They either keep them on hold or send them generic template messages which don't give applicants proper feedback and insights into their performance.

There are many digital platforms to cater to the recruitment process, but most of them focus on only one aspect of the entire process. This results in the recruiters juggling multiple platforms making the process ineffective and time taking. Furthermore, there are many advancements in artificial intelligence and machine learning and prompt engineering is the most recent advancement in that, yet most applications fail to use them effectively thus not providing advanced features like resume parsing, summary generation of each step and prompt-based interactions which can significantly enhance the recruitment process.

2.4 Goals and Objectives

The primary goals and objectives of this project are:

- Create an intuitive and user-friendly web application that streamlines the different aspects of the recruitment process in one place
- Utilize the abilities of LLMs and evaluate the effectiveness of prompt engineering to optimize different stages of the recruitment process.
- Automate resume parsing and filtering based on user-defined criteria.
- Optimize interview feedback and analysis to generate comprehensive summaries that offer deep insights into applicants' and interviewers' performance.
- Enable effective interaction between applicants and recruiters ensuring timely feedback and clear guidance on the next steps.

2.5 Project Scope

The scope of the Recruit Ease is centered around the development of a user-friendly web-based application intended to automate, streamline and optimize various aspects of the recruitment process using

advanced technologies like LLMs and prompt engineering. The application will enable the users to:

- Signup, Create and Manage profiles
- Upload, read and delete resumes in bulk
- Set specific job descriptions against job titles that will be used for candidate filtering
- Perform resume screening based on company rules and job descriptions
- Effectively see the candidate profiles while adding the interview feedback against some predefined criteria
- Receive intelligent summaries after every step of the hiring process
- Automate communications with candidates, including effective feedback delivery

The project scope does not include hosting or server infrastructure for the application. Neither does it include mobile application or device responsiveness. Furthermore, integration with third-party recruitment platforms and external APIs is out of scope.

The application will be built using modern web-development frameworks like React.js coupled with additional libraries and tools for styling. The back end will be developed using a framework like Node.js or Django. We plan to utilize two databases: a vectorDB to store the LLM responses and a PostgreSQL or MongoDB to store the application data. The application will leverage LLMs and prompt engineering and for this, we will be using LangChain. We will be using Agile Model as our project management methodology.

The major deliverable of the project will be a functional Web Application that will help organizations and recruiters to improve the effectiveness of their recruitment process. Furthermore, there would be a comprehensive user manual and detailed documentation on the literature, requirement specification, design, development, implementation and testing phases of the project.

2.6 Sustainable Development Goal (SDG)

Recruit Ease aligns with the Sustainable Development Goal of “decent work and economic growth”. By optimizing the hiring process and ensuring the qualified and deserving candidates are not overlooked, the application promotes sustained and sustainable economic growth. This provides a platform that ensures efficiency in talent acquisition, thus everyone has a chance for productive employment and decent work.

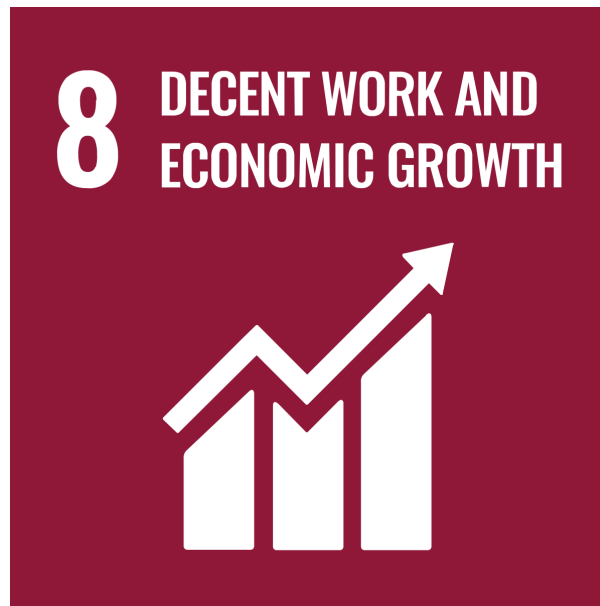


Figure 2.1: Sustainable Development Goal

This figure represents our sustainable development goal which is decent work and economic growth

2.7 Constraints

For a system like Recruit Ease to be introduced, there are the following constraints that need to be brought under consideration:

- The effectiveness of the application greatly depends on the capabilities of the LLMs. If there are limitations in the LLMs ability to understand and parse certain information, it could impact the overall efficiency of the application.
- The traditional HR departments might be reluctant to change or be skeptical about automating the recruitment process.

2.8 Business Opportunity

As the landscape of recruitment evolves, the application provides a significant business opportunity. Revenue can be generated through the following:

- Memberships can be offered to larger companies which could include large corporations and other small and medium sized companies.
- Some advanced features could be introduced or insights available for premium members
- As the platform's user base grows, advertisements could be incorporated.
- Recruitment trend reports and market analysis can be made available for sale based on the platform

data.

2.9 Stakeholders Description

The system will be used by the following:

- **System Administrator:**

These are the people that will oversee and maintain the overall functionality and health of the platform. They can add or remove user profiles while managing the platform settings and addressing technical issues.

- **Recruiters (HR department):**

These are the main users that will utilize the platform to optimize the different stages of the recruitment process. They will have the ability to upload Resumes, get them parsed, set job descriptions and criteria, generate summaries and communicate with the applicants.

- **Interviewers:**

These are the people that will be using the application to add the interview feedback against. They will have the ability to see the complete profile of the candidates they will be interviewing and add feedback and a final verdict after the interview.

- **Organizations:**

These are the larger entities or corporations looking for an effective and streamlined recruitment process. They will have the ability to purchase the application, get special features added to them and benefit from a centralized recruitment process.

2.9.1 Stakeholders Summary

The targeted impact of Recruit Ease can be achieved by the collaboration and endorsement from multiple stakeholders. Firstly the System Administrator ensures the platform's effectiveness and seamless performance by maintaining and refining the application based on the evolving needs. Secondly, the Recruiters or the HR departments will drive most of the application's functionalities and they effectively determine the application's overall success. Next, the interviewer's interaction with the application ensures that the platform's insights are accurate and actionable. Lastly, the Organizations bring vision to scale. When these large entities opt for Recruit Ease, they not only validate its effectiveness but also promote it as a trusted and reliable solution in the recruitment industry. The synergy between all these stakeholders ensures that Recruit Ease remains a reliable, effective, optimal and highly sought-after solution in the recruitment landscape.

2.9.2 Key High-Level Goals and Problems of Stakeholders

The primary objective of the application is to revolutionize the recruitment process and automate and optimize the various stages of talent acquisition. By offering one platform for all processes, the goal is to simplify the recruitment process for HR departments, interviewers and organizations. While this seems to be a good solution, there are some problems as well. A problem for the stakeholders could be to adapt to this digital shift, especially for those that are accustomed to traditional recruitment methods. Additionally the system administrators might face a problem of persuading diverse organizations to adopt this novel approach. Lastly, the platform's success hinges on trust, which would require the assurance about the accuracy of LLM-driven data analysis and data privacy.

2.10 Conclusion

Recruit ease is an application that aims on transforming the way people are hired in companies. It is a web-based application that harnesses the capabilities of LLMs and prompt engineering to streamline the different stages of the recruitment process. The project vision is to revolutionize the recruitment industry by improving the entire process along with the communication between employers and applicants. The main goal of the application is to create a user-friendly platform that makes hiring easier and efficient. However, there are some potential challenges like exploring this field of technology and making it work for our particular use case along with training people on using it. This application offers huge opportunities for companies by simplifying the recruitment process for them.

Chapter 3 Literature Review / Related Work

This section includes detailed literature review related to the automated recruitment process. We have explored how artificial intelligence is shaping HR management, the best prompting techniques used to get precise answers from LLMs, and the technologies that boost the power of these language models.

3.1 Definitions, Acronyms, and Abbreviations

ATS: Application Tracking System

CoT: Chain of Thought

CoT-SC: Chain of Thought Self-consistency

e-HRM: Electronic Human Resource Management

HR: Human Resource

HRIS: Human Resource Information System

LLM: Large Language Models

LSTM: long short-term memory

NLG: Natural Language Generation

NLI: Natural Language Inference

NLP: Natural Language Processing

NLU: Natural Language Understanding

OOD tasks: Out of Domain tasks

TELeR: Turn, Expression, Level of Details, Role

VDBMS: Vector Database Management Systems

3.2 Detailed Literature Review

The following sections provide each review material in detail including its summary, the critical analysis and relation to our work.

3.2.1 Zoho Recruit

This article signifies the importance of automated HR systems and discusses some of the most common and important features in ATS by taking a use case in consideration. This takes Zoho Recruit as a case study and discusses the major functionalities provided by it. [1].

3.2.1.1 Summary of the research item

HR is not just one function, today it consists of a set of highly specialized capabilities. Organizations have realized its significance and thus have started moving from traditional systems to new and IT inspired HR systems also referred to as e-HRM(electronic human resource management). Application Tracking Systems are emerging systems in the broader e-HRM. Just the way IT influenced almost every field, it has changed the recruitment method as well, the traditional methods of recruitments included job postings in magazines, newspapers etc while the modern methods include the online job posting.

The basic features of HRIS generally include various features including management of employee information, reporting and analysis of that information, basic company related documents concerning employees and their policies, integration with the payroll system and most importantly application tracking and resume management system. Zoho Recruit is a web application designed to streamline the different steps of the recruitment process. Zoho recruit offers features like candidate sourcing, resume parsing, resume management and custom requirement workflows. Its major features are that it reduces the recruitment costs and time. Furthermore, it also enhances the internal and external communications. This system caters to the staffing agencies and the corporate HRs.

3.2.1.2 Critical analysis of the research item

The research article provides a comprehensive discussion of the transition from traditional to modern HR systems that are very relevant in today's rapidly changing, IT influenced era. Zoho Recruit provides an all-encompassing perspective, offering a complete overview of clients, contacts, candidates, and job requisitions. Furthermore, it efficiently automates the recruitment process and offers its users with customization. It has smooth integration with other third-party apps and is also cost effective as compared to other ATS in the market. However, it is limited for hard-copy resumes and every candidate must be associated with a job opening. It has the potential to fall for inaccurate keyword matching and that the outdated job postings remain active.

3.2.1.3 Relationship to the proposed research work

The research article provides a good understanding of how the recruitment process and HR systems have revolutionized from the traditional method to modern technologies. It discusses e-HRMs and our project is a direct enhancement in the evolution that is discussed in the article. While Zoho recruit offers a good solution for recruitment, Recruit Ease aims to solve the issues using the power of LLMs. Recruit Ease can learn from the strengths of Zoho recruit, while it can address the weaknesses of the application

3.2.2 RecruitPro: A Pre Trained Language Model with Skill-Aware Prompt Learning for Intelligent Recruitment

The research article is a collaborative work by multiple scholars. Together, these authors present a unique approach to intelligent recruitment, emphasizing skill-aware prompt learning and its implications in refining the recruitment process through advanced language models [2].

3.2.2.1 Summary of the research item

RecruitPro emerges as a revolutionary solution in the world of smart recruitment. In recent times, the surge in machine-learning-driven recruitment services has become evident. There has been a lot of progress with tools that excel in areas such as job classification, salary forecasting, and assessing the relationship between job and individual, but the overwhelming majority remain tailored to specific domains or tasks. This specific focus can limit their flexibility, particularly when they can cater to a variety of industries due to constraints in their training data.

To bridge this gap, the paper introduces RecruitPro, a flexible framework developed with a unique skill-aware, prompt-based pre-training approach. The authors took an extended approach by first introducing a model embedded with context that was first tested on a large recruitment sample. To assess its effectiveness, they've developed 13 benchmark tasks that all represent different stages of the recruiting process.

RecruitPro's framework is based on skill-aware prompt learning, which is central to its design. This module increases the pre-trained model's flexibility, ensuring that it is not only versatile but also precise. RecruitPro is in essence a holistic tool, aiming to seamlessly link the various aspects of recruiting, indicating a promising direction for future advancements in this sector.

3.2.2.2 Critical analysis of the research item

"RecruitPro" successfully overcomes the limitations of current machine-learning recruitment tools by aiming for greater flexibility. The foundation of a large recruitment database is promising, but the choice of just 13 benchmark tasks may limit its real-world application. The use of skill-aware prompt learning is exciting, but its viability in an ever-evolving career landscape is uncertain. Drawing parallels with existing methods would raise its credibility. RecruitPro, in essence, carries a lot of promise, but a thorough analysis in real-world contexts is essential to assess its adaptability.

3.2.2.3 Relationship to the proposed research work

Our proposed project involves the creation of a web-based RecruitEase Application based on prompt engineering and LLM, The findings and methods of RecruitPro can be directly applicable. the RecruitPro-style prompt learning approach, which is based on skill awareness, could greatly improve the RecruitEase Application's efficiency and precision. Understanding the tools used in RecruitPro and their potential drawbacks can provide insight into improving the application's parsing, classification, and matching algorithms. RecruitEase may also benefit from adopting the concept of unified representation, as discussed in the report, in terms of flexibility to serve diverse industries and professional positions.

3.2.3 Applicant Tracking Software and Hiring Platform: Green House

This review evaluates Greenhouse Software which is a leading recruitment platform that streamlines hiring processes, enhancing transparency and candidate experience. With powerful tools and analytics, it assists companies in finding top talent efficiently [3].

3.2.3.1 Summary of the research item

Greenhouse is an ATS designed to optimize the entire recruitment process for organizations ranging from small size to enterprise level. It offers multiple features to streamline the hiring process. Firstly, it provides a structured hiring approach for consistent and standardized hiring to ensure alignment between recruiters and hiring managers. To aid this it provides dashboards and reports for teams to track each aspect of the recruiting process efficiently. Moreover, Green House is also beneficial in talent sourcing as it provides interactive career pages which help in attracting potential candidates for job positions. With the help of personalized and interactive features it also enhances candidate experience leading to a better perception of the organization. Green House ensures Diversity, Equity and Inclusion (DEI) by focusing on tools and strategies to implement diverse hiring and an inclusive recruitment process. Furthermore, Green Houses is known for continuously updating its features and implementing new solutions to better suit the evolving needs of businesses and their recruitment teams.

3.2.3.2 Critical analysis of the research item

Greenhouse stands out as one of the leading ATS platforms offering a combination of interactive user interface and advanced features to help in the recruitment process. Its overall feedback from its user base tends to be positive. However, potential adopters must be aware of certain aspects like pricing which is not transparent and is known to be higher than others making it unsuitable for small to medium sized organizations.

3.2.3.3 Relationship to the proposed research work

Greenhouse's platform includes features for streamlining the recruitment process. The methodologies they have adopted could offer insights, best practices, and innovative approaches that could be valuable to our project. By studying Greenhouse's features, analytics, user experience, we could potentially identify the mandatory features and user expectations for our own system.

3.2.4 Artificial Intelligence Chatbots are New Recruiters

This research article assesses the ability of chatbots in enhancing the recruitment process. In the modern era of recruitment, chatbots have emerged as indispensable tools, reshaping and enhancing the way organizations interact with potential employees. These AI powered systems not only streamline administrative tasks but also significantly improve the candidate experience [4].

3.2.4.1 Summary of the research item

Modern chatbots accelerate the recruitment process in a number of ways. They simplify the initial filtering process by interacting with applicants and asking them general questions. Information gathered is used for quickly shortlisting potential candidates. They also ensure accurate and timely data collection by regularly updating candidate databases. Furthermore, chatbots excel in distinguishing suitable candidates by asking questions related to skills, qualifications, and experiences. This frees the recruiters from these time taking and exhaustive tasks. Furthermore, they help in administrative duties like scheduling interviews and informing candidates about their performance and next hiring steps. Chatbots have enhanced application rates by engaging potential candidates effectively on messaging platforms and handling their FAQs. This increases the likelihood of application completion as chatbots remind them of deadlines and respond to queries based on keyword recognition, making the process more user-friendly. Moreover, chatbots play a significant role in maintaining positive candidate perceptions. They update applicants about their application status, reducing anxiety and wait times. Furthermore, chatbots offer constructive feedback even in rejection scenarios. This not only preserves a positive company image but also opens doors for potential future engagements.

3.2.4.2 Critical analysis of the research item

The article offers a detailed and comprehensive overview of the integration of chatbots in the hiring process. It rightly emphasizes the transformative role of chatbots in helping recruiters and enhancing candidate experience, citing real-world examples. However, the narrative could be enriched with quantitative data to further substantiate its claims. Additionally, a more balanced perspective that addresses

potential limitations or challenges posed by chatbots, such as biases or industry-specific efficacies, would add depth.

3.2.4.3 Relationship to the proposed research work

Drawing from the comprehensive analysis of the article, our project aims to leverage the transformative potential of chatbots, particularly in streamlining recruitment processes, enhancing candidate experience, and ensuring timely and constructive feedback. The strengths outlined in the article, such as proactive FAQ handling, immediate candidate engagement, and the fostering of positive candidate perceptions, serve as foundational pillars for our automated HR recruiter. Integrating these insights, our project aspires to harness the full potential of AI-powered chatbots, setting new standards in HR recruitment and candidate engagement.

3.2.5 A Survey on Evaluation of Large Language Models in Natural Language Processing Tasks

The article centers on understanding the progression and evaluation of Natural Language Processing (NLP) tasks in Large Language Models [5].

3.2.5.1 Summary of the research item

This research article assesses strengths and vulnerabilities of LLMs in various NLP tasks including Natural Language Understanding (NLU), Reasoning, Natural Language Generation (NLG), Multilingual tasks and Factuality.

NLU includes sentiment analysis, text classification, natural language inference (NLI) and semantic understanding. LLMs have shown commendable performance in sentiment analysis with chatGPT being able to interpret the emotional inclination of the text in binary and triple class classification problems. LLMs also showed acceptable accuracies over 85% in text classification. In NLI, where the goal is to assess if a hypothesis logically stems from a given premise, research indicates that Large Language Models (LLMs) have considerable potential for further development. Semantic understanding remains a challenge for LLMs, particularly in evaluating basic phrases and distinguishing between meaningful and nonsensical phrases. GPT-3.5 and Bard consistently classified highly nonsense phrases as meaningful. GPT-4 showed significant improvements, but its performance is still significantly lower than that of humans.

The reasoning ability of Large Language Models (LLMs) span areas like mathematical, commonsense, logical, and domain-specific reasoning. ChatGPT excels in arithmetic reasoning compared to GPT-3.5 but falls short in broader mathematical reasoning and symbolic tasks. Furthermore, ChatGPT and

GPT-4 outdo traditional methods in logical reasoning but struggle with new and out-of-distribution data. Nonetheless, while LLMs present a progressive trend in reasoning tasks, they continue to face challenges, emphasizing the need for further research and development.

NLG evaluates LLMs in activities like creating summaries, creating dialogues, translation, and answering questions. In summarization, TNLG v2 and OPT outperformed others, but ChatGPT sometimes creates summaries lengthier than the source. For dialogue tasks, both Claude and ChatGPT performed superiorly to GPT-3.5, with Claude slightly edging out in specific configurations. Regarding machine translation, ChatGPT and GPT-4 exhibited strong performances, although improvements are needed in English to non-English translations. In question-answering, InstructGPT davinci v2 was top-tier, with LLMs showing promising capabilities in other general knowledge areas. In essence, while LLMs have made significant strides in NLG tasks, there's still potential for refinement.

While many LLMs are trained on mixed-language data, they've primarily been evaluated on English data, neglecting multilingual performance assessment. Recently, several studies have revealed gaps in their multilingual capabilities. Notably, ChatGPT showed reduced performance in standard Arabic tasks and struggled with non-Latin languages. The findings emphasize the need for future research to prioritize multilingual balance, ensuring a more globally inclusive and unbiased approach.

Factuality in LLMs pertains to the model's alignment with verifiable facts and real-world truths. It's crucial for tasks like question-answering, information extraction, and automated fact-checking where misinformation can lead to significant misinterpretations. Several studies examined major models like GPT-4 and found they answer correctly over 80% of the time but still fall short of complete accuracy. Efforts also include breaking down LLM-generated text into atomic facts and introducing datasets like TruthfulQA to test model truthfulness. The collective findings underscore the ongoing challenge of ensuring factuality and the need for continued advancements in evaluation methods.

3.2.5.2 Critical analysis of the research item

The article offers a comprehensive exploration of key aspects of LLMs, emphasizing their capabilities in various NLP tasks. The coverage of multiple recent studies provides a broad overview of the current state of the field, highlighting various limitations and approaches to improve LLMs. While the article provides a detailed evaluation of LLMs, it might lean heavily on specific models and datasets, potentially neglecting a broader spectrum of LLM applications or other lesser-known models.

3.2.5.3 Relationship to the proposed research work

The NLP tasks evaluated in the article are relevant to our automated recruitment project. Firstly, NLU and NLG tasks will equip us with an understanding of how LLMs generate and process textual content,

which will be essential in parsing, analyzing and summarizing candidates resumes and interview feedback. The section on reasoning offers insights into how we can make more informed decisions or match candidates to suitable roles. Most critically, the importance of factuality as discussed ensures that the app remains accurate and trustworthy in its outputs, reducing potential hiring biases or errors. Altogether, this literature review provides a holistic overview of the functionalities and challenges associated with LLMs, which can be instrumental in shaping the design, efficiency, and effectiveness of our recruitment app.

3.2.6 Attention Is All You Need

The article introduces a new network architecture known as the "Transformer," which uses only attention mechanisms, removing the requirement for recurrence and convolutions in sequence transduction models. [6].

3.2.6.1 Summary of the research item

Traditionally, recurrent neural networks, especially long short-term memory and gated recurrent neural networks, are considered vital for sequence modeling tasks like language modeling and machine translation. However, a major drawback of recurrent models is their inherently sequential nature which makes it challenging to parallelize operations during training, especially for long sequences. On the other hand, attention mechanisms, which enable models focusing on specific aspects of the input for prediction, have emerged as essential components in sequence modeling. They efficiently model dependencies irrespective of the distance between elements in sequences.

The Transformer architecture proposed in this article relies entirely on a mechanism called self-attention, to represent relationships between different positions within a sequence. Similar to numerous neural sequence transduction models, the Transformer utilizes an encoder-decoder framework. This structure involves the encoder transforming a sequence of input symbols into continuous representations. Subsequently, the decoder produces an output sequence of symbols, where each step is influenced by the symbols generated earlier. The Transformer employs layers of self-attention and point-wise, fully connected networks for both its encoder and decoder components. Residual connections and layer normalization are used around sub-layers, ensuring consistent output dimensions.

This architecture has demonstrated success across multiple tasks, such as reading comprehension, generating summaries, textual entailment, and the development of task-independent sentences. Through experiments on machine translation tasks, the Transformer outperformed existing models in translation quality specifically, in the WMT 2014 tasks for translating English into German and English into French. The researchers express optimism regarding the use of attention-based models in different fields

and intend to investigate their application in various tasks and modalities.

3.2.6.2 Critical analysis of the research item

The article presents in an elaborative way the main strengths of the Transformer model's ability to train more efficiently than its recurrent or convolutional counterparts. Its structure, built entirely on attention mechanisms, showcases a potential paradigm shift in how neural architectures might be conceived for sequence-based tasks. However, the model has potential drawbacks. Since it relies completely on attention, it can be computationally intensive for tasks with long sequences. Additionally, Transformer excels in translation but its adaptability and performance across more wide range of tasks need to be thoroughly investigated. Furthermore, model relies heavily on data and computational power, which could be limiting resources. To summarize, while the Transformer presents a compelling new approach, its broader implications and potential limitations need to be further explored.

3.2.6.3 Relationship to the proposed research work

Transformer architecture is highly significant to the development of an automated HR recruiter app that leverages language models. The traditional use of recurrent neural networks is inherently sequential, thus potentially slowing down the processing of extensive datasets, such as large pools of resumes or lengthy job descriptions. The Transformer's advantage in parallelizing operations can lead to quicker and more efficient candidate screening, vital for real-time HR tasks. Its reliance on self-attention ensures the model can capture intricate details and dependencies in candidate profiles, possibly leading to more accurate matching of candidates to job descriptions. The demonstrated success of the Transformer in tasks like reading comprehension and abstractive summarization directly correlates with the demands of an HR application.

3.2.7 TELeR: A General Taxonomy of LLM Prompts for Benchmarking Complex Tasks

This article introduces TELeR taxonomy to categorize and standardize prompts for LLMs in complex tasks. By defining the dimensions: Turn, Expression, Role, Level of details, this approach provides a framework for researchers to standardize the prompts and perform meaningful comparisons [7].

3.2.7.1 Summary of the research item

The recent conversational LLMs like GPT-3, LLaMA, BArd, PaLM and BLOOM have shown outstanding effectiveness across a broad spectrum of basic NLP tasks, but when it comes to the performance in complex tasks, the styles of prompting plays a significant role which needs to be studied. A complex task is a task that involves a set of subtasks that require a high level of content and semantic understanding

along with planning and reasoning. The author explains that there are several key factors that can result in varying performance and accuracy of LLMs for complex tasks. Initially, the Level of Detail about the task to be performed in the prompt plays a significant role as a vague prompt can quite easily confuse the LLM. Additionally, clearly specifying the goal or the output required from the LLM can help achieve better results. When the data to be used by the LLM is user provided and the LLM shouldn't use its pre-defined knowledge, explicitly specifying that the associated data set is provided can change the effectiveness of the LLM. When performing a complex task having multiple subtasks, clearly specifying them in bullets helps the LLM generate better results, and then asking for reasoning or justification of the result helps make the output valuable for the user. Additionally, supplying a few examples in a few-shot manner and explicitly defining the context and role in the prompt aids the model by offering supplementary information for generating precise responses. The style of expression and interaction also significantly influences the accuracy and effectiveness of prompts in yielding the intended outputs. The proposed taxonomy, TELeR, proposes to categorize the LLM prompts along four dimensions: Turn, Expression, Role and Level of Details. The prompt may be structured as either single-turn or multi-turn based on the task's demands and complexity. Also, the manner of expression used in the overall task and its subtasks can greatly affect the outcomes. Furthermore, defining the role along with the level at which the details are specified can create a difference in the accuracy of the result produced. Lastly, two practical use-cases illustrate the applicability of the TELeR taxonomy.

3.2.7.2 Critical analysis of the research item

The research article addresses a gap in the LLM research which is about categorizing prompts for complex tasks and proposes a taxonomy, the TELeR taxonomy, to fill in this gap. Furthermore, the article provides clear definitions and explanations for the four dimensions of the TELeR taxonomy, making it easier for researchers and further elaborates on this using two practical real-life use cases. However, there are certain weaknesses evident in the research paper because it doesn't provide clear empirical results or case studies to support its claim. Further the potential challenges or shortcomings of the TELeR taxonomy have not been discussed in this research article. Furthermore, the article claims the importance of studying prompt engineering for complex tasks and does not support its argument with the support of any evidence or studies.

3.2.7.3 Relationship to the proposed research work

In the proposed HR system, most of the tasks from initial resume screening and filtering to smooth interaction and communication between recruiter and candidate would be based on prompt engineering. Considering the very basic and initial task of giving the LLM a set of unstructured data and fetching

results from it of whether the candidate would be a suitable fit for the job or not is a complex task and it could be solved easily using the proposed TELeR taxonomy along with all the other complex tasks involved in the process. The provided information in the article about the level of details provided, the context specification, the output requirements etc can be used to improve the prompts and fetch the desired results from the LLM in a single shot. The TELeR taxonomy can help us achieve accurate, reliable and effective results for our complex tasks.

3.2.8 Chain-of-Thought Prompting Elicits Reasoning in Large Language Models

The research article explores two innovative approaches called chain-of-thought prompting and few-shot prompting to empower language models with enhanced reasoning abilities similar to humans [8].

3.2.8.1 Summary of the research item

Although increasing the size of language models has led to performance improvements in various domains, there remains a need for enhancements in challenging tasks such as arithmetic, common sense, and symbolic reasoning. These limitations can be addressed by combining two key ideas: chain-of-thought prompting and few-shot prompting. Chain-of-thought prompting implements a methodical approach, which involves decomposing a complex issue into a sequence of logical intermediate reasoning steps to find the final answer. This offers insights into the model's thought process, which can be crucial for understanding and debugging its behavior in case of any error. Few shot promoting involves the use of sample inputs and outputs. Combining both these techniques a model is provided with a prompt that consists of triples: input, chain of thought and output.

The authors shows the success of CoT prompting in various reasoning benchmarks of arithmetic, commonsense, and symbolic reasoning. In arithmetic reasoning a study was done on solving math word problems including GSM8K, SVAMP, ASDiv, AQuA, and MAWPS using five language models GPT-3, LaMDA, PaLM, UL2, and Codex. Key findings revealed that use of CoT was effective mainly with larger models (100B parameters) and offered more significant benefits for complex problems. Notably, this approach outperformed traditional methods and also appeared robust across different annotators and exemplar sets, demonstrating its potential as a generalized method for math word problem-solving. Furthermore, a study was done to evaluate the performance of CoT for common sense reasoning. A range of datasets including CSQA, StrategyQA, Date Understanding, Sports Understanding, and the SayCan was evaluated. In experiments, the PaLM language model with chain-of-thought prompting demonstrated enhanced performance, surpassing previous benchmarks on the StrategyQA and excelling in sports understanding more than a sports enthusiast. Furthermore, CoT prompting allowed language models to effectively tackle symbolic reasoning tasks such as last letter concatenation and coin flipping

scenarios. CoT proved crucial for models to not just handle tasks within their training scope, but to also generalize and apply learned concepts to out of domain (OOD) tasks. Combining chain-of-thought and few-shot prompting enhances the reasoning capabilities of large language models across arithmetic, commonsense, and symbolic tasks, facilitating improved performance and better generalization.

3.2.8.2 Critical analysis of the research item

The research article provides a comprehensive analysis of chain-of-thought prompting and few-shot prompting to enhance the reasoning abilities of LLMs. The method's effectiveness and broad potential has been demonstrated by conducting experiments on three major reasoning benchmarks: arithmetic, commonsense, and symbolic reasoning. Additionally, multiple, significant language models have been used for the research, providing a comprehensive view of how different models respond to the new prompting techniques. However, certain weaknesses are evident in the research. Firstly, CoT mimics human reasoning but there is no concrete evidence for that. Also, there's no guarantee of correct reasoning paths, which can sometimes lead to incorrect answers. This makes the reliability of the method questionable in critical applications. The chain-of-thought reasoning emerges mainly in large models, making it expensive for practical use. The article leaves open the need for further research in areas like inducing reasoning in smaller models, which suggests that the approach is still evolving and not fully optimized.

3.2.8.3 Relationship to the proposed research work

In an automated HR recruiter, every task from assessing a candidate's resume, to determining the fit between the candidate's skills and job requirements, requires a sophisticated level of reasoning. CoT would be significantly used to ensure that the automated HR system is not just keyword matching but genuinely understanding and reasoning through the information. Additionally, valuable insights into the model's thought process can aid in debugging and refining the system when it makes incorrect assessments. Furthermore, as recruitment needs change with time and job descriptions evolve, it's crucial for an automated system to generalize its learning to newer scenarios, like the OOD tasks. The combination of chain-of-thought and few-shot prompting can ensure that the system remains accurate, efficient and adaptable with changing HR landscapes.

3.2.9 ReAct: Synergizing Reasoning and Acting in Language Models

This seminal paper, "ReAct: Synergizing Reasoning and Acting in Language Models" from the Department of Computer Science at Princeton University is one of the most referenced papers in the world of large language models. The research introduces the ReAct methodology, infusing reasoning and acting

in LLMs as a solution to many major problems that language models face [9].

3.2.9.1 Summary of the research item

The ReAct method aims to imitate the human way of combining task-oriented actions with verbal reasoning, or inner speech that lets them rapidly acquire new tasks and execute effective decision-making in the face of uncertainties. By combining reasoning and acting with large language models, the ReAct method is designed to address a variety of language reasoning and decision-making tasks.

One of the said such problems is that even properly prompted LLMs are vulnerable to fact hallucinations and error propagation due to the reason that the CoT model is like a static black box which limits its ability to reason reactively. Other approaches of pre-trained models are also limited by their incapability to not maintain a working memory. Keeping these issues in mind, the ReAct approach ensures that the model gives answers, explains the underlying reasoning, and engages in interactions by referencing both its internal and external knowledge reservoirs. With a solid evaluation on esteemed datasets like HotpotQA and Fever, the paper affirms the practicality of the ReAct mechanism and its thought-action-observation process.

The paper claims that by allowing the language model to control the asynchronous movement of thoughts and actions, it makes it simple and straightforward to implement, general and extensible, performant and robust, and also human aligned and controllable. The authors support this claim by applying ReAct to the HotpotQA and Fever datasets. They selected a few cases from the training sets and constructed ReAct-format trajectories as few-shot examples. These trajectories consist of multiple thought-action-observation steps. The thoughts were multifaceted, including: decomposition of questions, information extraction from Wikipedia, commonsense and arithmetic reasoning, search reformulation, and answer synthesis. Several baselines were constructed by ablating different components of ReAct trajectories. These included standard prompting, chain-of-thought prompting (CoT), acting-only prompt, and others. The study also introduced a technique to combine internal (from the model itself) and external knowledge (e.g. Wikipedia). This involved switching between the ReAct and another method called CoT-SC, depending on various conditions related to the confidence or lack thereof in the answers generated. The results of these knowledge intensive reasoning tasks was that the ReAct method consistently outperformed the Acting-only approach and showed superior performance on the Fever dataset but slightly trailed CoT on HotpotQA. However the most promising results were displayed when combining ReAct and CoT-SC.

The paper also touches upon various related works, some of which include: CoT, WebGPT and SayCan and Inner Monologue. It also suggests room for improvement by expanding ReAct through multi-task training and integrating it with supplementary approaches such as reinforcement learning. This approach

could make the agent stronger and unlock more potential for Large Language Models (LLMs) in various applications.

3.2.9.2 Critical analysis of the research item

Taking a critical look at the ReAct approach reveals both its success and shortcomings. Its primary strength is its grounded approach, which ensures the authenticity of the proposed responses based on reliable data sources and solid reasoning. The method's versatility is good, as shown by its ability to seamlessly switch between ReAct and CoT-SC modes depending on the situation. ReAct has demonstrated to be superior to its predecessors in benchmark tests. However, it's not without its challenges. The comprehensive structure that is ReAct's strength can also lead to potential lapses in reasoning. Moreover, the intricate nature of its reasoning scheme may not suit all application environments, particularly those that require quick, straightforward answers.

3.2.9.3 Relationship to the proposed research work

The ReAct approach will greatly benefit our proposed HR recruitment web app, which is anchored in prompt engineering and LLM-powered capabilities. tasks such as filtering resumes and elucidating reasons for candidate selection can be greatly enhanced. This would translate to decisions that aren't merely data-driven but are intricately reasoned. The ReAct methodology can further be utilized for tracking and evaluating candidate and interviewer performance, ensuring insights are derived not just from raw data, but also from a holistic understanding of the context. As feedback mechanisms are integrated, the dynamic nature of ReAct can help in refining the app's algorithms, setting the stage for a recruitment tool that's both advanced and insightful.

3.2.10 Reducing Retraining by Recycling Parameter-Efficient Prompts

The article explores the strategies for Prompt Recycling in Large Language Models, which enables a prompt trained on one model to work on a new or an updated model without the need of retraining. It highlights the possibilities for enhancing the flexibility and efficiency of applying LLMs to various tasks [10].

3.2.10.1 Summary of the research item

When we talk about Large Language Models, one of the things that come under consideration is to formulate the strategies to optimize the use of task specific soft prompts, making the model multifunctional. Prompt Recycling is a concept that these strategies revolve around, and it is a conceptual framework where a prompt is initially trained on a source model and then it is carefully adapted for use with a new

target model. We need to have this and learn this because the existing models keep on updating and with every update, their response to our predefined prompts change.

The strategies allow us the adaption of prompts without relying to supervised pairs of prompts, task-specific data, or additional training updates with the target model, thereby rendering the proposed strategies groundbreaking. These suggest a reduction in the overall cost and resources that is conventionally needed in the case of re-tuning prompts for every model update. They allow the deployment of a single pre-trained LLM to conduct an array of tasks, optimizing model behavior by concatenating to the input text thus the implications of such strategies are considerable. The article demonstrates the feasibility of prompt recycling with a significant success rate of 88.9%. However the room for enhancing the performance and reliability of these recycling techniques is substantial and it further suggests that while the initial results are promising, the refinement and innovation in recycling methodologies are imperative for realizing the full potential of parameter-efficient prompts in LLMs.

3.2.10.2 Critical analysis of the research item

The approach presented in the article is an innovative approach to utilize the LLMs with the concept of Prompt Recycling which allows the enhancement of flexibility and resource efficiency by reducing the need for extensive retraining and task-specific data. The approach is amazing because of its application in various use cases and ensures that the integrity and functionality of models remain existent over time and is not compromised. However, despite the rate of 88.9% success in prompt recycling, there is still a clear need of further refinement and improvement. There is a dependency on the advancement in recycling techniques which implies that there are potential challenges in real-world implementation of the suggested approach. The challenges occur especially in maintaining the models accuracy and efficiency, necessitating ongoing developments to fully realize the practical applicability and reliability of the proposed methodologies.

3.2.10.3 Relationship to the proposed research work

The reviews research article is crucial for the proposed work Recruit Ease since it provides insights into optimizing the use of LLMs and prompting, which are the core components for the proposed project. Furthermore, the idea of prompt recycling could play an instrumental role in maintaining and updating the system and the existing prompts, ensuring that it stays relevant and that we are able to get our desired results despite the models being updated from time to time. This interrelation signals a potential pathway for the project to address challenges of model dependency and the need for continuous refinements. This would allow a more streamlined and efficient implementation of the automated recruitment process.

3.2.11 Creating Large Language Model Applications Utilizing LangChain: A Primer on Developing LLM Apps Fast

This research article was written by Oguzhan Topsakal and Tahir Cetin Akinci from Florida Polytechnic University and University of California at Riverside, respectively. This paper provides an overview of applications and critical features of Langchain [11].

3.2.11.1 Summary of the research item

The applications of LLMs within the current software landscape are thoroughly explored in the provided research study. Particular emphasis is placed on the utilization of LangChain, an open-source framework designed to foster the rapid development of LLM-centric applications. The authors convey that a growing number of individuals and organizations are harnessing this technology across diverse domains, including education, research, customer service, content creation, healthcare, and entertainment, among others. LangChain is presented as a pivotal solution to address the needs of these applications. To enhance comprehension, the study employs a structured approach. Initially, it explicitly defines the core components of LangChain, which include Prompts, Models, Chains, Memory, Question Answering for Documents, and Agents. This structured presentation establishes the basis for further clarification, where the study offers practical examples of how LangChain can be beneficial in different situations. This process enhances our comprehension of the real-world implications of LLMs and the significant role that LangChain plays in accelerating software development driven by LLMs.

3.2.11.2 Critical analysis of the research item

The document successfully provides a comprehensive overview of the LangChain library, highlighting its utility in the swift development of LLM-based applications. The modular nature of LangChain, with its components and chains, offers flexibility and customization, catering to specific use-cases. Practical examples elucidate the potential applications and versatility of the LangChain framework, demonstrating its real-world applicability. However, from the initial content, the depth of the study's empirical evaluations or comparisons with other existing tools is unclear. Also due to the fact that the paper primarily focuses on LangChain, which might limit its insights into the broader ecosystem of LLM application development tools.

3.2.11.3 Relationship to the proposed research work

In the context of creating an automated HR recruitment system, it's crucial to understand how these systems work. LangChain's flexible approach, which focuses on customizable processes, provides a

useful way to design complex decision-making capabilities within the HR system. This method ensures that the system goes beyond simple keyword matching and evolves to better understand and make sense of data. During the development phase, it's essential to keep the HR system adaptable, cost-effective, and able to respond to changes in the recruitment landscape. The insights from this report can guide the research efforts aimed at achieving these goals.

3.2.12 Vector database management systems: Fundamental concepts, use-cases, and current challenges

The research article, written by Toni Taipalus, focuses on vector database management systems, which have become crucial in contemporary data management. Rapid growth has been fueled by the increasing demand for computational analysis of complex data types, including texts, images, and videos, in fields such as recommendation systems, similarity search, and chatbots. [12].

3.2.12.1 Summary of the research item

The paper presents an introduction to Vector Database Management Systems (VDBMS), which are widely known in modern data management. VDBMS have been used in various areas such as recommender systems and chatbots due to the need for a computational representation of complex data structures, such as images, and videos. They are then converted into numerical vectors, which are cost-effective to store and comparing vectorized data's unique characteristics, particularly its high dimensionality and sparsity, necessitating specific storage, retrieval, and processing techniques. The article further explains the process of vectorization, wherein unstructured data is converted into vectors, capturing essential features. Such vectors can range from simple to complex attributes, determined by neural networks, with each dimension representing different characteristics. The vectorized data can then be queried to find similarities, with applications in areas like reverse image search. Efficient management of such data necessitates a dedicated VDBMS, which provides functionalities like transaction control, scalability, and query optimization. This study aims to offer a comprehensive understanding of VDBMS, from foundational concepts to current challenges, without delving into the intricacies of specific products.

A core application of VDBMSs is similarity search, which spans across different media. The article highlights the role of VDBMSs in enhancing chatbot memory. Generative models, underpinning many chatbots, often face recall limitations. VDBMSs serve as a repository, storing past interactions to provide chatbots with contextually aware and personalized responses to users.

VDBMSs, which search approximate nearest neighbors, grapple with balancing speed and accuracy. While some index types prioritize speed and storage efficiency, others focus on accuracy. The challenge

is the increasing dimensionality of vectors. As vectors expand to capture more features, storage needs grow, and traditional similarity measures become less effective. High-dimensional vectors also tend to be sparse, complicating their indexing and retrieval. VDBMSs, being relatively new, are on the path to maturity. As the demand for vector databases surges, providers are pushed to innovate, making vectors more interpretable and introducing efficient model fine-tuning methods.

3.2.12.2 Critical analysis of the research item

The article does a good job of breaking down VDBMS in a way that's easy to understand. It uses real-world examples to help explain the more theoretical points, which helps readers understand better. However, the article could use more hands-on examples to showcase how VDBMS works in real life. Also, while the article talks about the importance of turning data into vectors, it could explain more clearly why this is beneficial in different situations.

3.2.12.3 Relationship to the proposed research work

For projects like our HR Recruitment app that deal with sorting and matching data, the insights from this article are very useful. The article's discussion on changing unstructured data into organized, vector forms is similar to how we process resumes. The best practices and potential issues talked about in the article can help us make our app's backend better, ensuring fast and correct resume matches and suggestions. In essence, the article lays a theoretical foundation that can significantly inform the technical development of data-centric applications like the HR Recruitment app.

3.3 Literature Review Summary Table

This section contains a compact summary of the literature reviews we carried out. First table is of the various similar applications we reviewed. Second table summarizes the research articles.

Table 3.1: Summary of Similar Applications

This table provides summary of several HR applications similar to Recruit Ease

Application	Features	Relevance to Application	Limitations
Zoho Recruit [1]	Features like candidate sourcing, resume parsing, custom workflows and seamless integration	Automation and Refinement in the recruitment process	Challenged with hard-copy resumes, potential outdated job postings and lack of personal touch
RecruitPro [2]	Skill-aware prompt learning; Unified representation learning	Enhances resume parsing, job matching, and cross-industry adaptability	Restricted to 13 benchmark tasks; Longevity in evolving job markets untested
Green House [3]	Structured hiring, Talent sourcing, Candidate experience Diversity, equity and inclusion	Methodologies adopted could offer insights, best practices, innovative approaches and user expectations	Pricing is high

Table 3.2: Summary of Literature Review*This table provides summary of various related research papers*

Author	Method	Results	Limitations
N.Nawaz and A. M. Gomes,[4]	AI Chatbots Checked for their ability to summarize text, question and answer based on keywords	Acceptable performance in administrative tasks and interaction with applicants	Lack of quantitative data
Y.Chang, X.Wang, J.Wang, Y.Wu, K.Zhu, H.Chen, L.Yang, X.Yi, C.Wang, Y.Wang, et al. [5]	Natural Language Understanding, Reasoning, Generation and Multilingual tasks on LLMs	Acceptable parsing, analyzing and summarizing of textual data	Leans heavily on specific models and data sets
A.Vaswani, N.Shazeer, N.Parmar, J.Uszkoreit, L.Jones, A.N.Gomez, Ł.Kaiser, and I.Polosukhin. [6]	Transformer attention mechanism	Outperformed existing models in translation quality specifically, WMT 2014 English-to-German and English-to-French translation tasks	Not applicable with constrained resources
J.Wei, X.Wang, D.Schuurmans, M.Bosma, F.Xia, E.Chi, Q.V. Le, D.Zhou, et al. [7]	TELeR taxonomy divides prompts in four dimensions: Turn, Expression, Role, and Level of Details	The application of this taxonomy shows how these four dimensions can influence the performance of LLMs	Lack of potential challenges or shortcomings of TELeR taxonomy
S.K.K.Santu and D.Feng. [8]	Arithmetic, commonsense, and symbolic reasoning on LLMs using CoT technique	CoT was effective with larger models (100B parameters). Useful for both domain specific and OOD tasks	Reliability not ensured in critical applications, expensive as only in large models
S.Yao, J.Zhao, D.Yu, N.Du, I.Shafran, K.Narasimhan, and Y.Cao. [9]	Large-scale benchmarking using ReAct on data sets like HotpotQA and Fever	Superior performance with ReAct; Enhanced when combined with CoT-SC	Slightly trailed CoT on HotpotQA; Potential reasoning lapses in complex scenarios
B.Lester, J.Yurtsever, S.Shakeri [10]	Prompt recycling allows a prompt trained on one LLM to be adapted on a new or updated LLM without retraining	Prompt recycling strategies demonstrated an 88.9 percent success rate	The initial results are promising but there is a need for refinement in recycling methodologies.
O.Topsakal and T.C.Akinci. [11]	Analysis of LangChain for LLM applications	LangChain's versatility in multiple domains; Defined core components	Limited evaluations; Narrow focus on LangChain
T.Taipalus. [12]	Examination of VDBMS and their applications	VDBMS's importance in data management; use in chatbots and similarity search	Needs more real-world examples; could delve deeper into vector benefits

3.4 Conclusion

To conclude the literature review covers three major aspects of LLMs including the NLP techniques that are used to build them, efficient prompting methods to generate answers and the technologies used to build applications using LLMs. We analyze the role of artificial intelligence, particularly chatbots in modern recruitment process stages like initial filtering, feedback generation and effective applicant communication. RecruitPro introduces a skill-aware prompt learning approach for intelligent hiring. Other papers explore the NLP and transformer approaches used in the advancements of LLMs. We have further explored multiple prompting approaches like CoT, few shot prompting, ReAct and Parameter-Efficient Prompts for the efficient utilization of LLMs. Lastly, papers on LangChain and vector database management systems explore the development of advanced tools used to build applications using LLMs.

Chapter 4 Software Requirement Specifications

This section provides an in-depth analysis of all associated software requirements of our project. It includes the list of features, the functional requirements, the quality attributes, the non-functional requirements, related assumption, the hardware and software requirements, the use cases, graphical user interface, all necessary diagrams and figures, along with the risk analysis of our project Recruit Ease.

4.1 List of Features

The following features will be available in the system An simple interface for HR recruiters to manage, assess, and interact with applicant resumes.

- An simple interface for HR recruiters to manage, assess, and interact with applicant resumes.
- Allows resumes to be uploaded in both PDF and DOC formats.
- Enables HR recruiters to manage and process job descriptions.
- Harness the capabilities of prompt engineering to automate the screening of resumes against specific job criteria and company policies.
- Provides a platform for interviewers to submit comprehensive feedback on shortlisted candidates.
- Creates an executive summary including the key aspects of an applicant's profile and interview feedback, aiding HRs in decision-making.
- Creates email drafts to convey rejection feedback or the next procedure to candidates in a gentle and constructive manner.

4.2 Functional Requirements

This section contains functional requirements for different stakeholders involving Admin, HR and Interviewer.

4.2.1 Functional Requirements for Admin

- The system shall allow the admin to log into the application using their login credentials
- The system shall allow the admin to sign up new users
- The system shall allow the admin to delete a user
- The system shall allow the admin to edit existing users

- The system shall allow the admin to create new user groups
- The system shall allow the admin to modify existing user group
- The system shall allow the admin to create new access permissions
- The system shall allow the admin to edit access permissions
- The system shall allow the admin to add permissions to a user group
- The system shall allow the admin to remove various permissions from a user group
- The system shall allow the admin to assign users to user groups
- The system shall allow the admin to remove users from a user group
- The system shall allow the admin to access all other features available in the application as well

4.2.2 Functional Requirements for HR

- The system shall allow the HR to log into the application using their credentials
- The system shall allow the HR to upload resumes for initial screening and further processing
- The system shall allow the HR to parse uploaded resumes to extract relevant details
- The system shall allow the HR to add new job descriptions
- The system shall allow the HR to view job descriptions
- The system shall allow the HR to edit existing job descriptions
- The system shall allow the HR to assign interviewers as collaborators for specific applicant profiles
- The system shall allow the HR to remove interviewers as collaborators from applicant profiles(s)
- The system shall allow the HR to view the interview feedbacks on applicant profiles
- The system shall allow the HR access to all applicant profiles
- The system shall allow the HR to generate executive summaries after every recruitment phase
- The system shall allow the HR to generate emails to be sent to the applicants

4.2.3 Functional Requirements for Interviewer

- The system shall allow the interviewer to login the application using their credentials
- The system shall enable the interviewer to submit feedback for assigned candidates
- The system shall let interviewers access profiles of their interviewed or assigned candidates.

4.2.4 Functional Requirements for System

- The system shall work smoothly with LangChain for effective data sharing and communication.

4.3 Quality Attributes

The project aims to incorporate the following fundamental quality attributes in order to ensure the prevention of application defects.

4.3.1 Reliability

The application must reliably process data and apply prompt engineering for accurate results. Furthermore, the systems should operate reliably ensuring that the data is not compromised or lost in any kind of application features.

4.3.2 Flexibility

The system shall be able to respond to uncertainty. It will adapt easily to the possible changes in the environment in the future and allow users to customize certain criteria to meet their specific needs.

4.3.3 Security

The system shall implement principles of security to help the software remain functional in response to the possible attacks. It should protect data from unauthorized access and breaches and ensure data confidentiality and integrity.

4.3.4 Maintainability

It must be easy to maintain and update the application thus, it should support updates to existing features and provide the support to add new features.

4.3.5 Usability

The system should provide such an interface and experience to the users that it is easy for them to navigate through the application. It shall provide a smooth user experience in response to their queries so the interaction could become seamless.

4.3.6 Performance

The system shall perform effectively when there are time and resource constraints.

4.3.7 Efficiency

The application should operate efficiently, even when there is a high-volume recruitment period going on. It should use the available resources in an optimized way.

4.4 Non-Functional Requirements

This section mentions several non-functional requirements to ensure performance and efficiency of the system.

4.4.1 Performance

- The system shall load data and display results in under 5 seconds.
- The system shall ensure a response time of less than 5 seconds.
- The system's bounce rate should remain between 20 to 35 percent.

4.4.2 Usability

- The system shall enable HR recruiters to upload a resume in no more than 4 steps.
- The system shall allow interviewers to submit feedback within 3 steps.
- The system shall let HRs view applicant profiles in 2 steps.
- The system shall be responsive, ensuring compatibility across different screen sizes.
- The user interface shall be intuitive and straightforward for users of all technical proficiencies.

4.4.3 Scalability

- The system shall support over 1000 job applicants.
- The system shall accommodate at least 200 users with different permissions and roles.
- The system shall provide a consistent and unified platform for developers and database administrators (DBAs) of the system.

4.4.4 Security

- The system shall safeguard all data transfers of the application.
- The system shall encrypt the passwords stored in the system's database.

4.5 Assumptions

No special assumptions are made at this point.

4.6 Use Cases

This section lists relevant use cases that represent central functionalities of our system encompassing all stakeholders.

4.6.1 Login

Name	Login		
Actors	Admin, HR, Interviewer		
Summary	The user shall provide their email and password on the login form, and after successful verification, redirect the user to the home page.		
Pre-Conditions	The user must be in the database records, either added by any of the authorized users or added manually by a developer. The user must not be logged in.		
Post-Conditions	The user’s session is successfully established and shall be redirected to the home page.		
Special Requirements	None		
Basic Flow			
Actor Action		System Response	
1	The user opens the login page.	2	The login page is displayed asking for email and password.
3	The user enters valid email and password.	4	The system verifies the email and password, establishes a session for the user and redirects the user to the home page.
Alternative Flow			
3	The user enters invalid email or password.	4-A	The system responds with an error message: Incorrect email or password entered.

4.6.2 Logout

Name		Logout	
Actors		Admin, HR, Interviewer	
Summary		The user shall click on the ”Logout” button and will be redirected to the login page.	
Pre-Conditions		The user must be be logged in.	
Post-Conditions		The user’s session is successfully ended and shall be redirected to the login page.	
Special Requirements		None	
Basic Flow			
Actor Action		System Response	
1	The user clicks on the ”Logout” button.	2	The system terminates the user’s session and redirects the user to the login page
No Alternative Flow			

4.6.3 Sign Up New Users

Name		Sign Up New Users	
Actors		Admin	
Summary		The User registers a new user in the system.	
Pre-Conditions		The User must be logged in to the system.	
Post-Conditions		The newly registered user’s profile is created and stored in the system’s database.	
Special Requirements		None	
Basic Flow			
Actor Action		System Response	
1	The Admin navigates to the "Create New User" section.	2	The "Create New User" Page is displayed.
3	The Admin fills out the required fields.	4	The System validates the entered data.
5	The Admin clicks the "Register" button.	6	The System creates a new user profile and stores it in the database and confirms the successful creation of the new user to the Admin
Alternative Flow			
3	The user enters invalid data.	4-A	The system responds with an error message: Incorrect data entered.

4.6.4 Edit User

Name		Edit User	
Actors		Admin	
Summary		The User edits an existing user in the system.	
Pre-Conditions		The User must be logged in to the system.	
Post-Conditions		The already existing user’s profile is updated in the database.	
Special Requirements		None	
Basic Flow			
Actor Action		System Response	
1	The Admin navigates to the ”All Users” section.	2	The ”All Users” page is displayed.
3	The Admin clicks the edit icon in front of an existing user section.	4	The User existing profile is displayed.
5	The user edits the required fields.	6	The System validates the entered data.
7	The user clicks the ”Edit” button.	8	The System edits the existing user profile and stores it in the database and confirms the successful update to the user
Alternative Flow			
5	The user enters invalid data.	6-A	The system responds with an error message: Incorrect data entered.

4.6.5 Delete User

Name		Delete User	
Actors		Admin	
Summary		The User deletes an existing user in the system.	
Pre-Conditions		The User must be logged in to the system.	
Post-Conditions		The existing user’s profile is deleted from the database.	
Special Requirements		None	
Basic Flow			
Actor Action		System Response	
1	The Admin navigates to the ”All Users” section.	2	The ”All Users” page is displayed.
3	The Admin clicks the delete icon in front of an existing user section.	4	A confirmation message to delete is displayed.
5	The user clicks the confirm option.	6	The System deletes the mentioned user from the database and confirms the successful update to the user
No Alternative Flow			

4.6.6 View All Users

Name		View All Users	
Actors		Admin	
Summary		The User shall access a comprehensive list of all users who are present in the system.	
Pre-Conditions		The user must already be logged in.	
Post-Conditions		The System shall display a list of all users, including their basic details and permissions assigned to them.	
Special Requirements		None	
Basic Flow			
Actor Action		System Response	
1	The User navigates to the "View All Users" section.	2	The System retrieves all the users data from the database and displays it.
No Alternative Flow			

4.6.7 Add New Job Description

Name		Add New Job Description	
Actors		HR, Admin	
Summary		The User shall introduce a new job description detailing the requirements, roles, and responsibilities for a specific position.	
Pre-Conditions		The user must already be logged in.	
Post-Conditions		The System shall have the newly added job description stored in its database and made available for relevant processes.	
Special Requirements		None	
Basic Flow			
Actor Action		System Response	
1	The User navigates to the "Add New Job Description" section	2	The "Add New Job Description" page is displayed.
3	The User fills out the necessary fields.	4	The System validates the entered data.
5	The User clicks the "Submit" button.	6	The System stores the new job description in the database and confirms the successful addition of the new job description to the User.
Alternative Flow			
3	The user enters invalid data.	4-A	The system responds with an error message: Incorrect data entered.

4.6.8 Edit Job Description

Name		Edit Job Description	
Actors		HR, Admin	
Summary		The User shall be able to edit an existing job description in the system.	
Pre-Conditions		The user must already be logged in.	
Post-Conditions		The System shall update the job description in database.	
Special Requirements		None	
Basic Flow			
Actor Action		System Response	
1	The User navigates to the "All Job Description" section	2	The "All Job Description" page is displayed.
3	The User selects the edit icon in front of the job description.	4	The System displays the existing job description data.
5	The User edits the necessary fields.	6	The System validates the entered data.
7	The User clicks the "Edit" button.	8	The System stores the updated job description in the database and confirms the successful update of the job description to the User.
Alternative Flow			
5	The user enters invalid data.	6-A	The system responds with an error message: Incorrect data entered.

4.6.9 Delete Job Description

Name		Delete Job Description	
Actors		HR, Admin	
Summary		The User shall be able to delete an existing job description from the database	
Pre-Conditions		The user must already be logged in.	
Post-Conditions		The System shall delete the specified job description from the database.	
Special Requirements		None	
Basic Flow			
Actor Action		System Response	
1	The User navigates to the "All Job Descriptions" page	2	The "All Job Description" page is displayed.
3	The User clicks the delete icon in front of a specific job description.	4	The System displays a confirmation pop up
5	The User clicks the "Confirm" button.	6	The System deletes the job description from the database.
No Alternative Flow			

4.6.10 View All Job Descriptions

Name		View all Job Description	
Actors		HR, Admin	
Summary		The User shall be able to edit an existing job description in the system.	
Pre-Conditions		The user must already be logged in.	
Post-Conditions		The System shall be able to view all job description in database.	
Special Requirements		None	
Basic Flow			
Actor Action		System Response	
1	The User navigates to the "All Job Description" section	2	The "All Job Description" page is displayed.
No Alternative Flow			

4.6.11 Upload Resumes

Name	Upload Resumes		
Actors	HR, Admin		
Summary	The User uploads resumes.		
Pre-Conditions	The User must be logged in to the system.		
Post-Conditions	The uploaded resumes are stored in the database		
Special Requirements	None		
Basic Flow			
Actor Action		System Response	
1	The Admin navigates to the "Upload Resumes" section.	2	The "Upload Resume" page is displayed.
3	The Admin clicks on Add button.	4	The System opens the tab to select documents from device.
5	The Admin selects the resumes	6	The System uploads all resume documents and stores them in the database and confirms the successful addition
Alternative Flow			
5	The user selects corrupted or non accessible file.	6-A	The system responds with an error message: File not accessible.

4.6.12 Initial Screening Process

Name		Initial Screening	
Actors		HR, Admin	
Summary		The User selects resumes and the respective job descriptions for running the initial screening.	
Pre-Conditions		The User must be logged in to the system.	
Post-Conditions		The initial screening results are stored in the database.	
Special Requirements		None	
Basic Flow			
Actor Action		System Response	
1	The Admin navigates to the "Initial Screening" section.	2	The "Initial Screening" page is displayed.
3	The Admin selects resumes and job descriptions.	4	The System runs the initial screening process displays it on screen and store the results in database.
Alternative Flow			
5	The user leaves either resumes or job description empty.	6-A	The system responds with an error message: Upload required data.

4.6.13 Add Interview Feedback

Name	Add Interview Feedback		
Actors	HR, Admin, Interviewer		
Summary	The User shall be able to add interview feed back for applicants		
Pre-Conditions	The user must already be logged in.		
Post-Conditions	The user shall be logged in and has the collaborator access to the applicant profile.		
Special Requirements	None		
Basic Flow			
Actor Action		System Response	
1	The User navigates to the "Individual Applicant" section.	2	The system displays the "Individual Applicant" profile.
3	The User uploads the feedback document or type the text and clicks "Submit" button.	4	The system shall generate summary and store it in the database.
No Alternative Flow			

4.6.14 Generate Email

Name		Generate Email	
Actors		HR	
Summary		The User shall request the system to generate emails.	
Pre-Conditions		The user must already be logged in.	
Post-Conditions		The System shall have the generated email ready for the user review and potential dispatch.	
Special Requirements		None	
Basic Flow			
Actor Action		System Response	
1	The User navigates to the "Generate Email" section.	2	The system displays the "Generate Email" page.
3	The User provides necessary inputs for the email.	5	The System, using the system, crafts the appropriate email based on the input or purpose.
4	The User clicks "Generate" button.	6	The System presents the generated email to the User for review.
Alternative Flow			
7	The User clicks "Regenerate" button.	8	The System presents the regenerated email to the User for review.

4.6.15 Generate Executive Summary

Name	Generate Executive Summary		
Actors	HR		
Summary	The User shall request the system to generate an executive summary.		
Pre-Conditions	The user must already be logged in. There should be an applicant in process for which the summary is being generated.		
Post-Conditions	The System shall provide the user with a detailed executive summary of the applicant’s progress and evaluation.		
Special Requirements	None		
Basic Flow			
Actor Action		System Response	
1	The User navigates to the "Generate Executive Summary" section.	2	The system displays the "Generate Executive Summary" page.
3	The User selects the specific applicant in process.	4	The System collates all relevant data and evaluations related to the selected applicant.
5	The User clicks "Generate" button.	6	The System generates the executive summary and presents it to the User for review.
Alternative Flow			
7	The User clicks "Regenerate" button.	8	The System presents the regenerated summary to the User for review.

4.6.16 View All Applicants

Name		View All Applicants	
Actors		HR, Admin	
Summary		The User shall access a comprehensive list of all applicants who are present in the system.	
Pre-Conditions		The user must already be logged in.	
Post-Conditions		The System shall display a list of all applicants, including their basic details and current status.	
Special Requirements		None	
Basic Flow			
Actor Action		System Response	
1	The User navigates to the "View All Applicants" section.	2	The System retrieves all the applicant data from the database and displays it.
No Alternative Flow			

4.6.17 View an Individual Applicant

Name		View an Individual Applicant	
Actors		HR, Admin, Interviewer	
Summary		The User shall access the detailed profile of a specific applicant	
Pre-Conditions		The user must already be logged in. The specific applicant’s profile should exist in the system.	
Post-Conditions		The System shall display the detailed profile, including all relevant details and evaluations of the selected applicant.	
Special Requirements		None	
Basic Flow			
Actor Action		System Response	
1	The User selects the desired applicant.	2	The System retrieves the detailed profile of the selected applicant from the database and displays it.
No Alternative Flow			

4.7 Hardware and Software Requirements

The hardware and software requirements of the application are mentioned below.

4.7.1 Hardware Requirements

We will need the following hardware components for the development:

- **Computer System:**

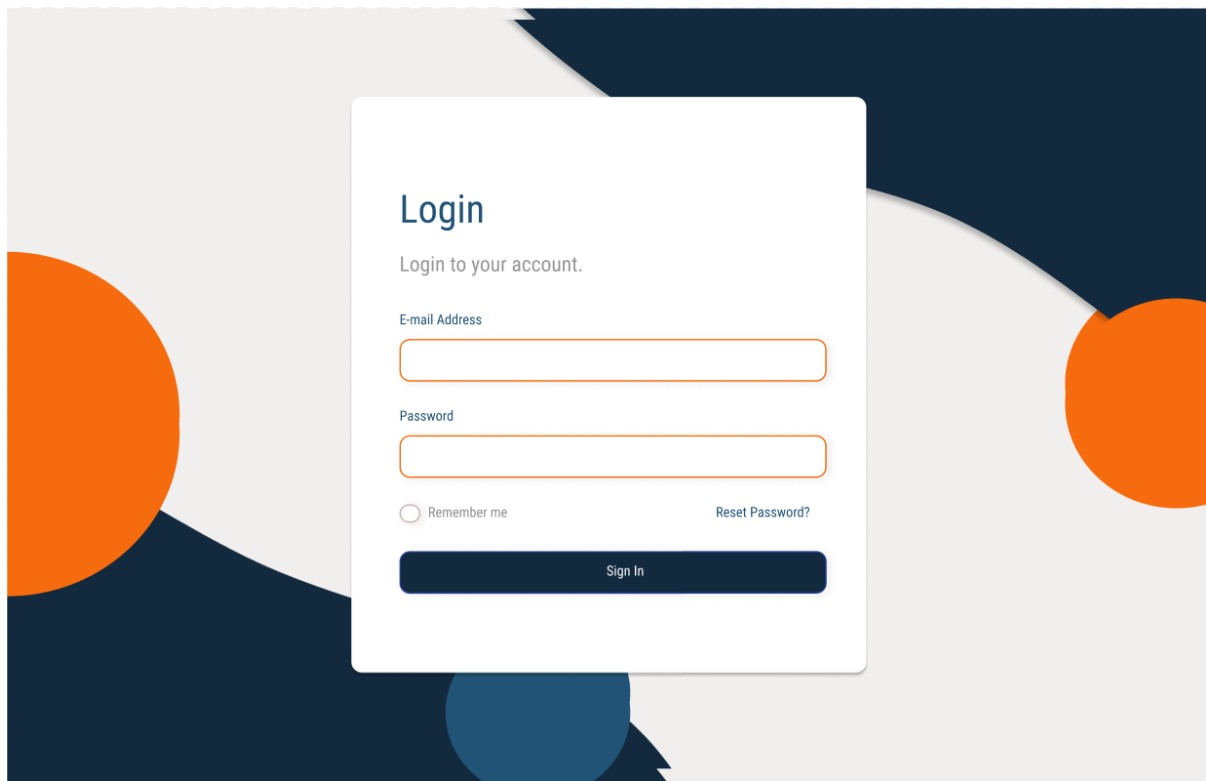
An internet capable computer system will be required for the development and testing of the project.

4.7.2 Software Requirements

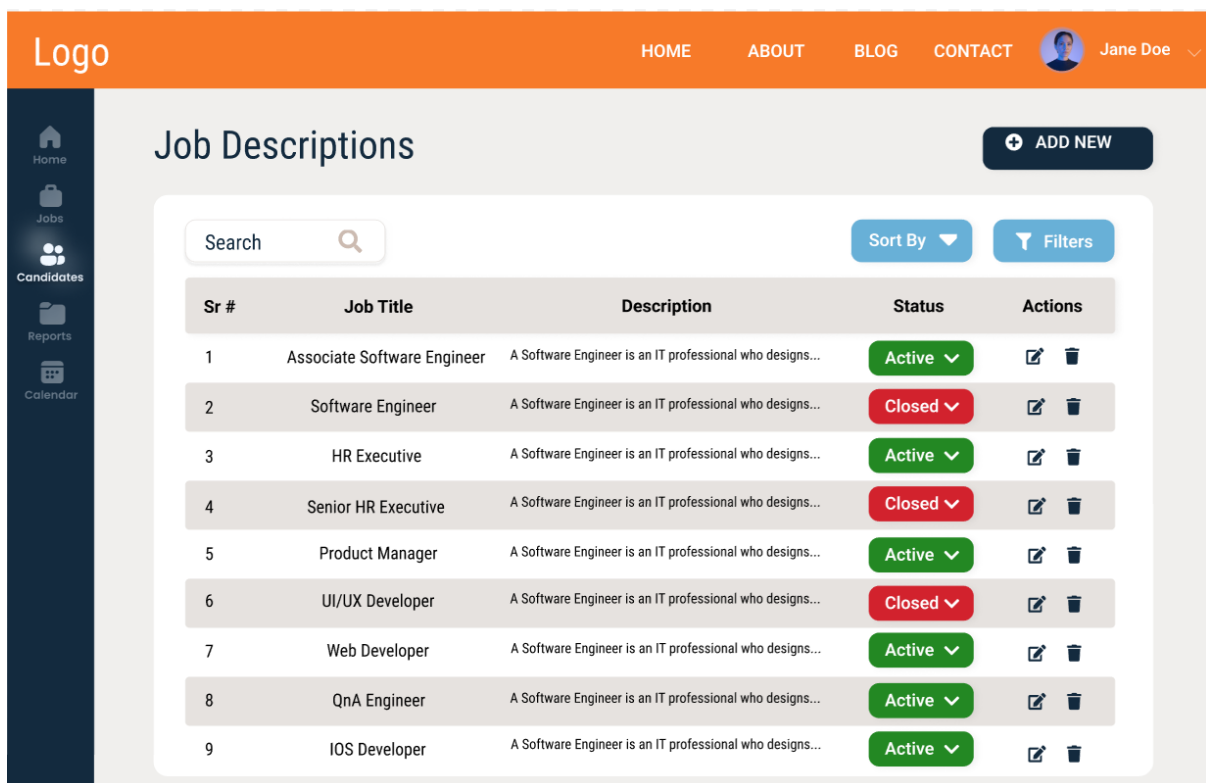
- **React JS:** JavaScript frontend library.
- **PostgreSQL Database:** SQL database.
- **Python:** For backend integration with Langchain.
- **Langchain:** Framework to simplify the creation of applications using LLMs (Large Language Models).
- **Django:** Python-based web-framework.
- **OpenAI API:** API key required to send requests to GPT for processing requests.
- **npm:** Node package manager for the JavaScript runtime environment of React.js.
- **Code Editor:** Integrated Development Environment (IDE) for development.
- **Tailwind CSS:** Open-source CSS framework.
- **Git:** Version control system.

4.8 Graphical User Interface

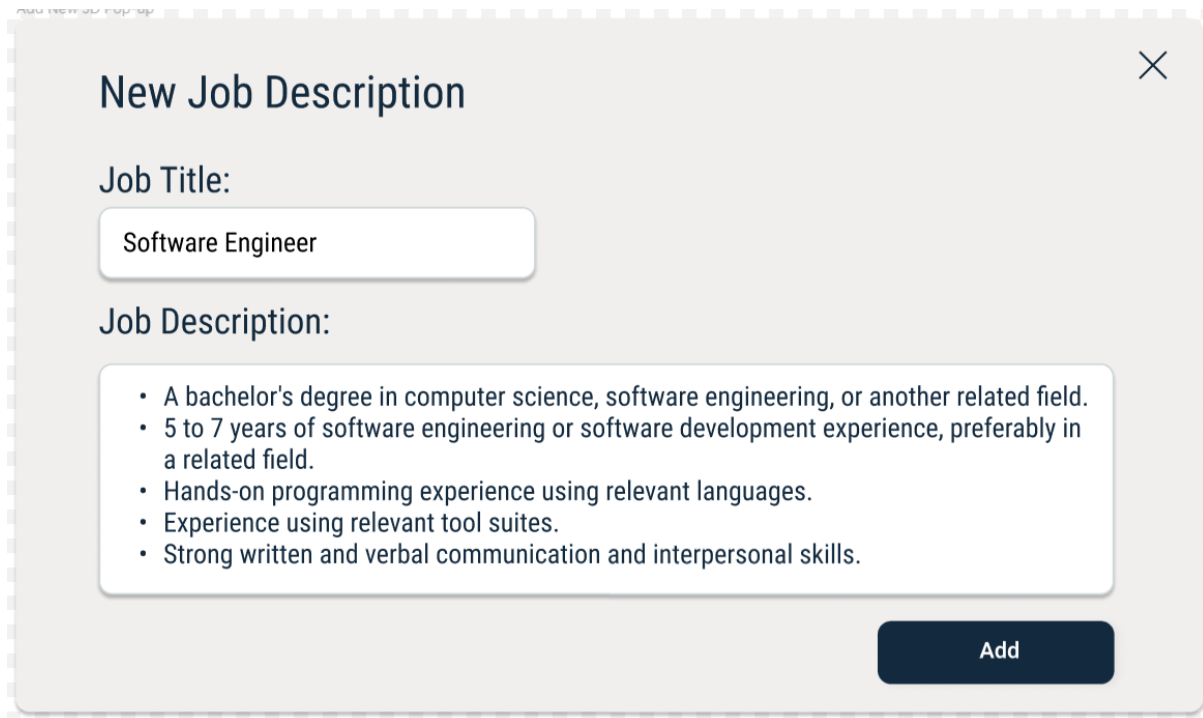
This section displays GUI screenshots of applications. The description of the GUI screens and the navigation flows are explained in the figure descriptions.

**Figure 4.1: Login Screen**

Login screen is displayed when the user wants to login to his portal. Login through email and password is the basic option.

**Figure 4.2: Job Descriptions Screen**

This screen shows all the job descriptions with their relevant information.



A modal popup titled "New Job Description" with a close button (X) in the top right corner. It contains two main sections: "Job Title:" with a text input field containing "Software Engineer", and "Job Description:" with a list of requirements. At the bottom right is a dark blue "Add" button.

New Job Description

Job Title:

Software Engineer

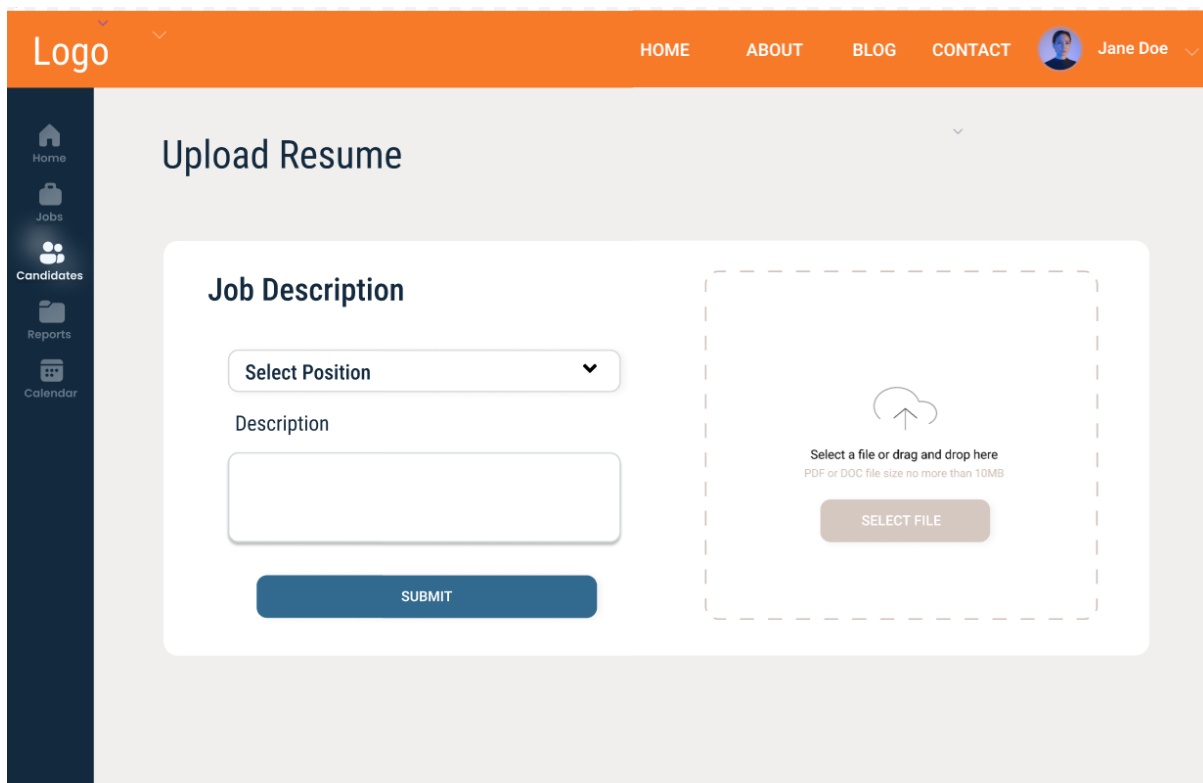
Job Description:

- A bachelor's degree in computer science, software engineering, or another related field.
- 5 to 7 years of software engineering or software development experience, preferably in a related field.
- Hands-on programming experience using relevant languages.
- Experience using relevant tool suites.
- Strong written and verbal communication and interpersonal skills.

Add

Figure 4.3: Add New Job Descriptions Popup

This popup shows up when user clicks on Add New button in the job descriptions screen.



A web application interface for uploading resumes. It features an orange top navigation bar with a logo, home, about, blog, and contact links, and a user profile for Jane Doe. A dark blue sidebar on the left contains icons for Home, Jobs, Candidates, Reports, and Calendar. The main content area is titled "Upload Resume" and contains a "Job Description" section with a position selector and a description text area, and a file upload section with a dashed box, upload icon, instructions, and a "SELECT FILE" button. A "SUBMIT" button is at the bottom of the job description section.

Logo

HOME ABOUT BLOG CONTACT Jane Doe

Home Jobs Candidates Reports Calendar

Upload Resume

Job Description

Select Position

Description

SELECT FILE

SUBMIT

Select a file or drag and drop here
PDF or DOC file size no more than 10MB

Figure 4.4: Upload Resume Screen

This screen shows is to upload resumes and select job description against which these resumes would be evaluated.

Results

Search Sort By ▼ Filters

Sr #	Name	Score	Description
☐ 1	Burak Bush	87%	the candidate seems to meet all the primary requirements of the JD and some o..
☒ 2	Obama Putseri	15%	the candidate seems to meet all the primary requirements of the JD and some o..
☐ 3	Candayce Wesley	10%	the candidate seems to meet all the primary requirements of the JD and some o..
☐ 4	John Dawis	20%	the candidate seems to meet all the primary requirements of the JD and some o..
☐ 5	Jordan Bush	90%	the candidate seems to meet all the primary requirements of the JD and some o..
☐ 6	John Hopkins	50%	the candidate seems to meet all the primary requirements of the JD and some o..
☐ 7	Emma Warson	89%	the candidate seems to meet all the primary requirements of the JD and some o..
☐ 8	Halston Brown	74%	the candidate seems to meet all the primary requirements of the JD and some o..
☐ 9	Taytum Fishman	23%	the candidate seems to meet all the primary requirements of the JD and some o..

Regenerate Proceed →

Figure 4.5: Screening Results Screen

This screen shows results of initial screening for all applicants with relevant information and actions.

All Applicants

Search Sort By ▼ Filters

Sr #	Name	Job Post Title	Date Applied/Modified	Status	Actions
1	Burak Bush	Associate Software Engineer	10/09/23	Shortlisted	
2	Obama Putseri	Associate Software Engineer	10/09/23	Interview Pending	
3	Candayce Wesley	HR Executive	10/09/23	Rejected	
4	John Dawis	Associate Software Engineer	10/09/23	Accepted	
5	Jordan Bush	Product Manager	10/09/23	On wait	
6	John Hopkins	Web Developer	10/09/23	Feedback Pending	
7	Emma Warson	Web Developer	10/09/23	Shortlisted	
8	Halston Brown	HR Executive	10/09/23	Interview Pending	
9	Taytum Fishman	IOS Developer	10/09/23	Shortlisted	

Figure 4.6: All Applicants Screen

This screen shows all the applicants with their relevant information.

Logo HOME ABOUT BLOG CONTACT Jane Doe

Home Jobs Candidates Reports Calendar

Kian Graham

KIAN GRAHAM
FLIGHT ATTENDANT

✉ kraham@company.com
📞 +1-555-123-4567
🌐 www.kiangraham.com
📄 View his full resume

RELEVANT SKILLS

- First Aid & Emergency Care
- Fluency in French
- Quality Customer Service

WORK EXPERIENCE

Flight Attendant
Winfield Airlines
March 2022 to August 2025 (3 years, 5 months)

- Boosted overall service quality by 35% by mentoring more than 40 junior flight attendants and training them in high-quality health and safety standards
- Maintained 100% positive feedback rating from passengers by providing them exceptional client-centered service and ensuring their comfort and safety
- Completed more than 3,000 hours of international and domestic flights

EDUCATION HISTORY

Bachelor of Science in Tourism
Institution: De Lougheigh University
Year of Graduation: 2020

- Cum Laude
- Best Thesis
- President, De Lougheigh Debate Club

CERTIFICATIONS

Flight Safety Training
Institution: National Aviation Association
2020

First Aid Certification
Institution: International Medical Agency
2020

1 Initial Screening

2 HR Interview

3 Technical Interview 1

4 Technical Interview 2

5 Culture Fit

Initial Screening

HR Interview

Technical Interview 1

Technical Interview 2

Culture Fit

Executive Summary

Generate Email

Figure 4.7: Individual Applicant Screen

This screen shows the entire recruitment process of an individual applicant.

Logo HOME ABOUT BLOG CONTACT Jane Doe

Home Jobs Candidates Reports Calendar

Interview Feedback

Kian Graham

Technical Interview - I
Software Engineer

Select a file or drag and drop here
DOC file size no more than 10MB

SELECT FILE

SUBMIT

Figure 4.8: Interview Feedback Screen

This screen allows users to upload interview feedback for an applicant.

Sr #	Name	UserName	Email	Role	Actions
1	Burak Bush	burak01	burak.bush@gmail.com	Super Admin	
2	Obama Putseri	obamaaaputseri	obama.putseri@gmail.com	HR Manager	
3	Candayce Wesley	wesleyycandaycee	candayce.wesley@gmail.com	Interviewer	
4	John Dawis	johndawwawis	john.dawis@gmail.com	Interviewer	
5	Jordan Bush	jordann178	jordan.bush@gmail.com	Admin	
6	John Hopkins	johnny	john.hopkins@gmail.com	Interviewer	
7	Emma Warson	watson.emma	emma.watson@gmail.com	HR Manager	
8	Halston Brown	brownnfam	halston.brown@gmail.com	HR Manager	
9	Taytum Fishman	twinmanfam	taytum.fishman@gmail.com	HR Manager	

Figure 4.9: All Users Screen

This screen shows all the users in the system with their relevant details.

Sr #	Group Name	No. Of Members	Policies Attached	Actions
1	Super Admin	2	13	
2	HR Manager	5	2	
3	Admin	3	7	
4	Interviewer	14	N/A	

Figure 4.10: User Groups Screen

This screen shows all user groups with relevant details.

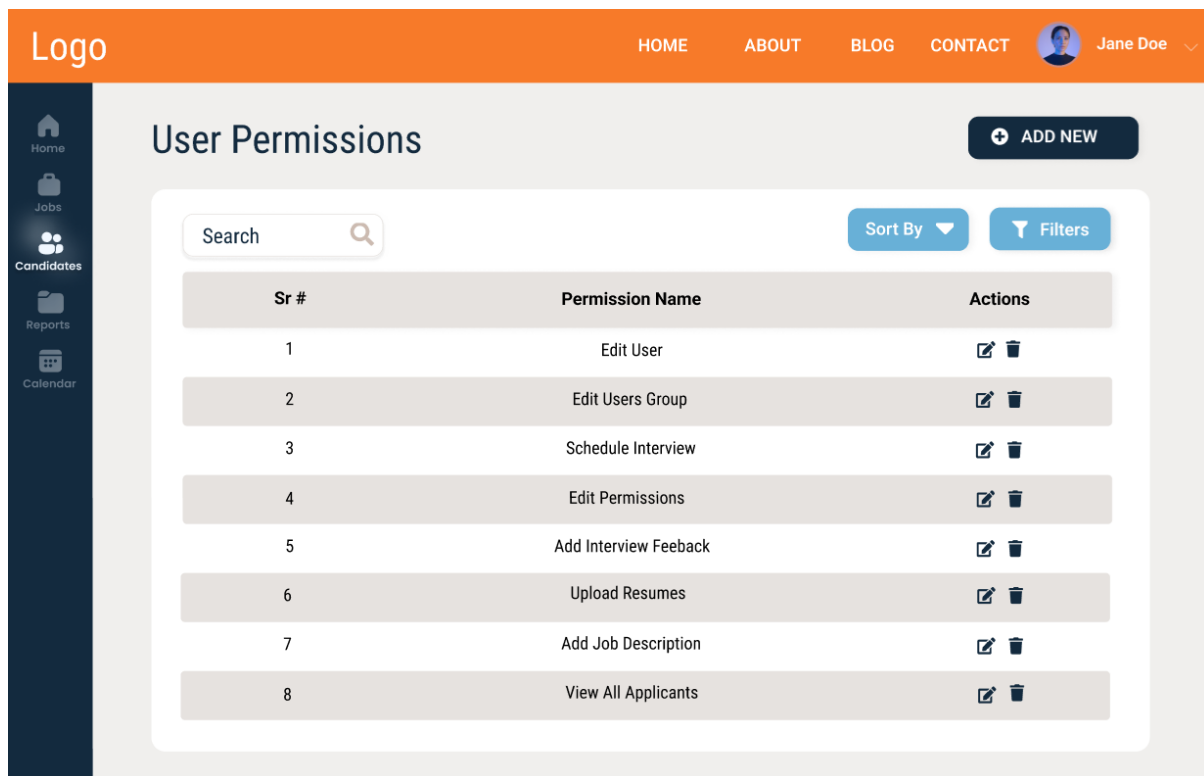


Figure 4.11: User Permissions Screen

This screen shows all user permissions based on major functionalities in the system.

4.9 Database Design

This section provides an overview of our database design. We have included an ER diagram and Data dictionary to show different classes, their attributes and relationships among them.

4.9.1 ER Diagram

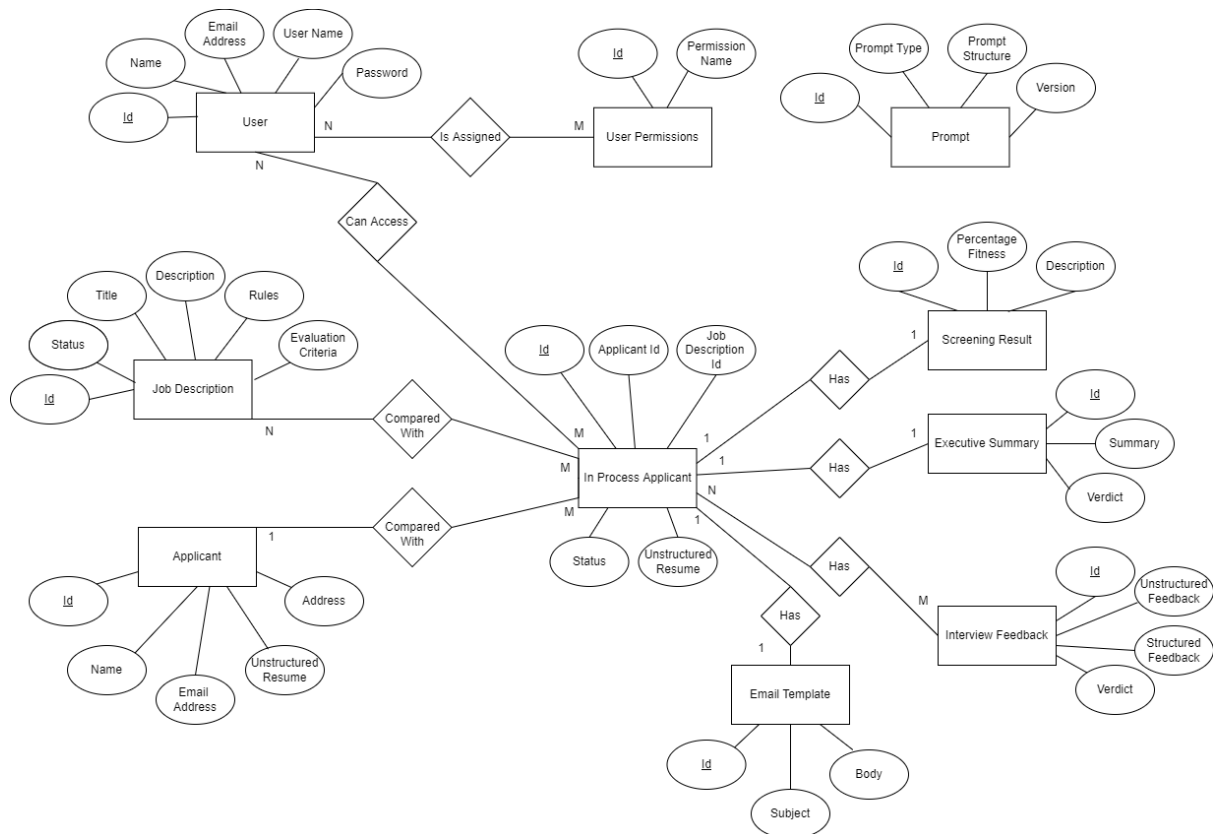


Figure 4.12: Entity Relationship Diagram

This figure represents all the entities and their relationship for the project in the form of an ER diagram

4.9.2 Data Dictionary

Table 4.1: Data Dictionary

This table represents the data dictionary of our system

Entity	Attribute	Data Type	Null-able	Relationship To	Relationship Type	Description
User	ID	Number	No	User Per- missions, In Process Applicant	N to M, N to M	ID of User (Primary key)
	NAME	Varchar(50)	No	-	-	Name of User
	EMAIL AD- DRESS	Varchar(50)	No	-	-	Email ad- dress of user

Continued on Next Page

Entity	Attribute	Data Type	Null-able	Relationship To	Relationship Type	Description
	USER NAME	Number	No	-	-	Unique username of User
	PASSWORD	Varchar(30)	No	-	-	Password of User
	DATE OF BIRTH	Date	No	-	-	Date of birth of User
User Permissions	ID	Number	No	User	M to N	ID of user (Foreign key)
	PERMISSION NAME	Varchar(30)	No	-	-	Permission name of user i.e Admin, HR or Interviewer
Job Description	ID	Number	No	In Process Applicant	N to M	Job Description ID (Primary key)
	TITLE	Varchar(100)	No	-	-	Title of the job description
	DESCRIPTION	Text	No	-	-	Description of the job
	RULES	JSON	Yes	-	-	Rules for the job (JSON format)
	EVALUATION CRITERIA	JSON	Yes	-	-	Evaluation criteria for the job (JSON format)

Continued on Next Page

Entity	Attribute	Data Type	Null-able	Relationship To	Relationship Type	Description
Applicant	ID	Number	No	In Process Applicant	1 to M	Applicant ID (Primary key)
	FULL NAME	Varchar(50)	No	-	-	Full name of the applicant
	CONTACT NUMBER	Varchar(11)	No	-	-	Contact number of the applicant
	EMAIL ADDRESS	Varchar(50)	No	-	-	Email address of the applicant
	ADDRESS	Varchar(200)	Yes	-	-	Address of the applicant
	UNSTRUCTURED DATA	File	No	-	-	Uploaded resume of the applicant
	STRUCTURED DATA	JSON	Yes	-	-	Structured data related to the applicant (JSON format)
Screening Result	ID	Number	No	In Process Applicant	1 to 1	Primary key
	PERCENTAGE FITNESS	Decimal(5,2)	No	-	-	Percentage fitness of the screening result
	SCREENING RESULT	JSON	Yes	-	-	Screening result in JSON format stored for application

Continued on Next Page

Entity	Attribute	Data Type	Null-able	Relationship To	Relationship Type	Description
Interview Feedback	ID	Number	No	In Process Applicant	M to N	Primary key
	UNSTRUCTURED FEEDBACK	Text	No	-	-	Unstructured feedback from the interview
	PREVIOUS STRUCTURED FEEDBACK	Text	No	-	-	Previous structured feedback from earlier rounds/interviews
	STRUCTURED FEEDBACK	Text	No	-	-	Structured feedback from the interview
	VERDICT	Varchar(50)	No	-	-	Verdict based on the interview feedback
Executive Summary	ID	Number	No	In Process Applicant	1 to 1	Primary key
	PREVIOUS SUMMARY	Text	Yes	-	-	Previous executive summary
	SUMMARY	Text	Yes	-	-	Current executive summary
	VERDICT	Varchar(50)	No	-	-	Verdict based on the executive summary

Continued on Next Page

Entity	Attribute	Data Type	Null-able	Relationship To	Relationship Type	Description
Email Template	ID	Number	No	In Process Applicant	1 to 1	Primary key
	SUBJECT	Varchar(200)	No	-	-	Subject of the email template
	PREVIOUS BODY	Text	Yes	-	-	Previous body/content of the email template
	BODY	Text	No	-	-	Current body/content of the email template
In Process Applicant	ID	Number	No	User, Job Description, Applicant, Screening Result, Executive Summary, Interview Feedback, Email Template	M to N, M to N, M to 1, 1 to 1, 1 to 1, N to M, 1 to 1	Primary key
	APPLICANT ID	Number (Foreign key)	No	-	Applicant	Reference to the Applicant
	JOB DESCRIPTION	Number (Foreign key)	No	-	Job Description	Reference to the Job Description

Continued on Next Page

Entity	Attribute	Data Type	Null-able	Relationship To	Relationship Type	Description
	STATUS	Varchar(50)	No	-	-	Status of the application (Using ApplicationStatus choices)
	SCREENING RESULT	Number (Foreign key)	Yes	-	Screening Result	Reference to the Screening Result
	INTERVIEWS	Number (Many-to-Many)	No	-	Interview Feedback	References to Interview Feedback
	EXECUTIVE SUMMARY	Number (Foreign key)	Yes	-	Executive Summary	Reference to the Executive Summary
	EMAIL TEMPLATE	Number (Foreign key)	Yes	-	Email Template	Reference to the Email Template
Prompt	ID	Number	No	-	-	Primary key
	PROMPT ID	Number	No	-	-	ID of the prompt (Primary key)
	PROMPT TYPE	Varchar(50)	No	-	Prompt Type	Type of the prompt (Using PromptType choices)
	PROMPT STRUCTURE	Text	No	-	-	Structure or format of the prompt

Continued on Next Page

Entity	Attribute	Data Type	Null-able	Relationship To	Relationship Type	Description
	VERSION	Positive Integer	No	-	-	Version number of the prompt

4.10 Risk Analysis

In this part we highlight some of the risks that are associated with the development, maintenance and scalability of the project.

4.10.1 Technical Risks

Working on very recent technology like prompt engineering, it is likely that there would be changes and advancements in it in the future that could possibly lead to the application requiring updates. Furthermore, working with APIs and development kits of the current age, it is likely that these will become obsolete in the future and have to be replaced. Furthermore, the requirements of the application could change in the future, thus asking for a product revamp.

4.10.2 Business Risks

The application might need financing in case it wants to host the application on the server or cloud. Additionally a need of developers and other resources might be felt in the future in order to maintain the application and scale up its scope to add other functionalities.

4.11 Conclusion

In conclusion, this chapter covers the project's Software Requirements specifications in details. It covers the necessary features, functional and non functional requirements along with the quality attributes of the application to make it an effective tool for the recruitment process. There is a special emphasis on the key characteristics of the application that it should be easy-to-use, flexible and secure. Use cases and use case description tables illustrate the system's functionalities and user interactions. This ensures a complete user-friendly recruitment application. The ER diagram and data dictionary included in the chapter provides a structured view of the system's data management and enhances the clarity and precision. The technology stack to be used for building the application is discussed. Furthermore, the potential risks from both the technological aspect and business aspect are discussed.

Chapter 5 Proposed Approach and Methodology

Our approach involves using a blend of new technologies to improve the HR recruitment process. A key component of our solution is the integration of OpenAI's Large Language Models (LLMs) with the Langchain library. Integration is critical for our system's powerful natural language processing (NLP) capabilities, which also enable our system to analyze text material with complex reasoning and better comprehension, which makes sure there is a high degree of precision in our recruiting processes.

5.1 OpenAI LLM

In our application, we plan on using OpenAI's Large Language Models (LLMs), particularly GPT3.5-Turbo and GPT 4. They respond well to Natural language queries that can be human-like responses to natural language queries due to their models. These models were trained on large datasets such as books, articles, and internet resources. Their ability to process and predict subsequent words in a conversation makes them useful in analyzing complex data in resumes and job descriptions.

LLMs help streamline the recruitment process and improve the assessment of candidates, which includes various tasks such as summarizing the candidate profile or extracting evaluation criteria from the Job Description. The choice between GPT-3.5-turbo and GPT-4 will depend on their respective performance metrics and cost-effectiveness.

5.2 Prompt Engineering

Understanding how prompt engineering guides Large Language Models (LLMs) toward producing good outputs is important. We design the prompt by integrating roles, context, inputs, and expected output format. The properly structured prompt would help us to make the LLM give consistent output, mimicking the way any other trained model would work.

While designing prompts, we cater to maintaining OpenAI's token limits. We also will design our application such that it integrates well with future advances. Furthermore, advances in OpenAI and LLM technologies would help in making our product more data-efficient and precise than traditional machine learning workflows,

5.3 Langchain

Langchain would be the most important tool in our application. It is used for utilizing LLMs and integrating them into user apps. It has various libraries such as prompt templates, parsers, chains, etc.

These libraries are pivotal in creating applications that are highly aware of the context. Using Langchain and its additional libraries we plan on making our application to interpret and give insightful results to our data i.e. Resumes, Job Descriptions, and Interview Feedback.

5.3.1 Pydantic Output Parser for Converting Text to JSON

We will use the Pydantic Output Parser to convert unstructured text data into a JSON object. This parser uses Pydantic base classes, which allows us to define the JSON structure we need for our data. This would ensure the consistency of the results of the LLM, which is critical for the effective process and analysis of the data in our application.

5.3.2 ChatModel for LLM Predictions

We will also use Langchains ChatModel, which allows for more conversational data processing. Using prompt templates with specific input and output variables, this method basically simulates real-world job interviews and scenarios. This capability is the core of our system, as it will allow the app to engage in interactive discussions with users that mimic real human interactions.

5.3.3 Assistants for Regeneration of Output and Prompt Fine-tuning

To regenerate the output and prompt fine-tuning, we would be using OpenAI assistants within the Langchain framework. These assistants follow instructions, execute multiple threads, and execute those with new inputs to improve the system's responses. It changes the output while considering user feedback, imitating the way a chatbox works. It would increase user satisfaction, and maintain and continually improve our recruitment system's flexibility and efficiency.

5.4 Conclusion

"Recruit Ease" aims to improve and refine the recruitment process by using these technologies. To make recruiting more effective and dynamic is based on the combined effectiveness of LLMs, the Langchain framework, and advanced prompt engineering techniques. It is a holistic approach that not only improves the efficiency of our recruiting processes but also guarantees a high degree of precision and flexibility in selecting candidates. Keeping our approach relevant and effective in the dynamic world of recruiting, this approach aligns well with the changing demands of the talent acquisition industry.

Chapter 6 High-Level and Low-Level Design

This chapter contains the High and Low level design of the system using various methods. This includes diagrammatic representations of the system through Sequence and Architecture Diagrams. It will also include an overview of the system and its different components.

6.1 System Overview

Recruit Ease is a web application designed to improve the recruitment process by integrating the capabilities of Large Language Models. We use prompt engineering to optimize efficiency, fairness, and precision in multiple recruitment stages.

6.1.1 Interactive User Interface

Our application will have an interactive user interface to provide a user-friendly experience. It will have dashboards which will be customized for different user roles like HR, Admin, Interviewer. They can view and manage activities according to their preferences. Furthermore, we will make our front end responsive to ensure that our application will be compatible across various devices.

6.1.2 Resume Parsing and Filtering

Our application will utilize LLMs to extract and interpret information from submitted resumes and job descriptions. Firstly, it will create a consistent structured format for every resume. Secondly, it will filter candidates based on job description. The entire process will ensure a fair and efficient screening process.

6.1.3 Interview Feedback and Analysis

Our application will automatically generate comprehensive summaries of interviews and various other hiring steps. These summaries will provide insights into candidates' performances. Performance metrics will be generated to help evaluate candidates and make informed decisions.

6.1.4 Applicant-Recruiter Interaction

Our application will facilitate seamless communication, providing timely feedback and guidance to applicants. Detailed feedback will be provided in all scenarios whether the candidate is accepted or rejected.

6.2 Design Considerations

This section describes many issues that should be considered before finalizing a complete design solution. These are essential to ensure that the end user does not face any problem while using the application.

6.2.1 Assumptions and Dependencies

The design of Recruit Ease is predicated on several assumptions and dependencies:

6.2.1.1 Related Software and Hardware

It is assumed that users will have access to basic computer hardware and stable internet connection. They would also be having stable versions of common browsers like Google Chrome/Mozilla Firefox or Safari in their devices.

6.2.1.2 Operating Systems

The application is designed to be compatible with major operating systems such as Windows, macOS, and Linux.

6.2.1.3 End-user Characteristics

Users are expected to have basic proficiency in web navigation and usage. It is also assumed that recruiters have basic knowledge of manual recruitment processes.

6.2.1.4 Possible and/or Probable Changes in Functionality

The field of recruitment is ever evolving with new job descriptions and requirements being added continuously. Thus, our application is designed to be adaptable to changing recruitment trends and technological advancements, especially in the field of LLMs.

6.2.2 General Constraints

Several global limitations and constraints have a significant impact on the design of Recruit Ease's software

6.2.2.1 Hardware or Software Environment

The system requires a stable internet connection and supported browser version. A good machine is necessary to ensure speed and efficiency. The application's performance is dependent on the consis-

tent availability of these resources and any fluctuations could affect the system's functionality and user experience.

6.2.2.2 End-user Environment

The application's usability depends on the end-user's familiarity with basic web applications.

6.2.2.3 Interoperability Requirements

The system needs to integrate seamlessly with the database and LLMs.

6.2.2.4 Interface/Protocol Requirements

Recruit Ease is designed to adhere to standard web protocols, including HTTP for data transmission. The interface is developed to be responsive and accessible, conforming to web standards like HTML5, CSS3, and JavaScript. This ensures cross-browser compatibility and a consistent user experience across different devices.

6.2.2.5 Data Repository and Distribution Requirements

In our application the data set would be huge as it would contain all resumes along with details of the entire recruitment process for each applicant. Thus, the application relies on a secure, scalable, and efficient database system for storing and retrieving data. The choice of database is crucial for handling large volumes of data and ensuring quick access.

6.2.2.6 Verification and Validation Requirements

Recruit Ease is an automated recruitment platform and it is highly important to ensure that there are no inaccuracies which can lead to unfair decisions. Apart from basic website functionalities, significant testing and refining is involved in case of LLMs and prompts also. Hence, extensive testing is necessary to ensure the application's reliability and efficiency.

6.2.3 Goals and Guidelines

The design and development of Recruit Ease are guided by several goals and guidelines. Firstly, we will follow the KISS principle which states that Keep It Simple, Stupid. This will ensure that the application remains straightforward and user-friendly, with no unnecessary complexity that could detract from user experience or system performance. Secondly, we will implement user-centric designs. This would involve creating an interactive interface that is understandable and accessible to all users. Recruit Ease design will cater to users with varying degrees of technology backgrounds. We will achieve this by

implementing easy navigation, and straightforward functionalities. Both the approaches will help to enhance the user experience.

6.2.4 Development Methods

The development of Recruit Ease is a combination of multiple different software development methodologies to ensure agility, adaptability, and quality in the final product. We will follow Agile method. This will allow iterative development, flexibility, and the ability to rapidly incorporate feedback into the development process. We will be able to respond quickly to changing requirements. Furthermore, we will use Scrum methodology to provide a structured and flexible way to manage the development process. Scrum will help in team collaboration, sprints, and progress meetings. Another important aspect of the development approach is Continuous Integration and Continuous Deployment. This method ensures that updates and improvements are seamlessly integrated and deployed without interrupting the user experience. All these methods will help ensure that the final product is technically sound, usable and fits the customer's needs.

6.3 System Architecture

Our application is designed to use modern applications such as Django, React, PostgreSQL, and Large Language Models (LLMs). The system's design includes both internal module architecture and external system interfaces. A diagrammatic representation is provided to help in understanding the system's structure.

6.3.1 High-Level Partitioning

The system is broken down into four main components

- **Presentation Layer:** In Front-end we would use React.js, for user interactions and session maintenance.
- **Business Logic Layer:** The backend will be built on Django, for implementing all of the application logic.
- **Data Persistence Layer:** PostgreSQL is used for data storage.
- **Integration with Large Language Models:** This layer would include the usage of OpenAI LLMs

6.3.2 Component Collaboration

All of the high-level components interact in a structured way. The Presentation Layer communicates with the Backend via RESTful APIs, sending and receiving data as required. Business logic Layer interacts with the Database for storage/retrieval and utilizes LLMs for enhanced data interpretation. The Database would be the central repository for all system data, maintaining integrity and security. LLM Integration provides analytical capabilities within the Backend, aiding the application's main use cases.

6.3.3 Rationale for Decomposition

The architecture allows for scalability, maintainability, and efficient handling of various functionalities. Alternative architectures were considered, but this one was selected for its ability to integrate LLMs effectively.

6.3.4 System Architecture Diagrams

The following diagrams are included for a better understanding of our system and their intricate relations.

6.3.4.1 High-Level System Architecture Diagram

The major layers and their primary functions are outlined in the high-level architecture diagram. The inner and outer flow of data and control is also highlighted.

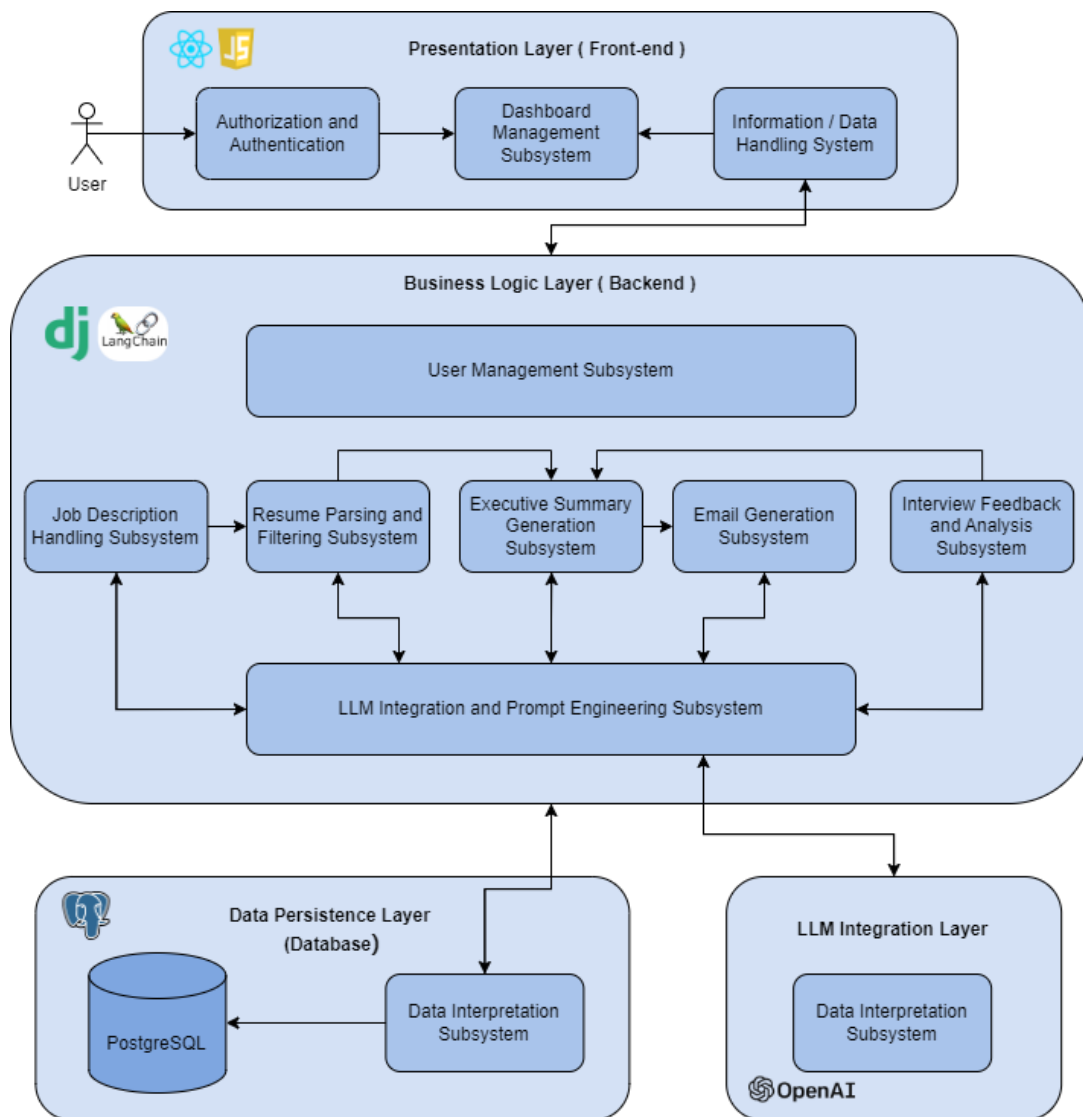


Figure 6.1: High-Level System Architecture Diagram

This diagram illustrates the high-level view of the system's architecture, major components, and their interactions.

6.3.4.2 Detailed Backend Services and LLM Integration Diagram

The diagram shows the inner workings of the services layer and its interaction with the integration layer, also focusing on the flow of information between the subsystems and the LLM layer.

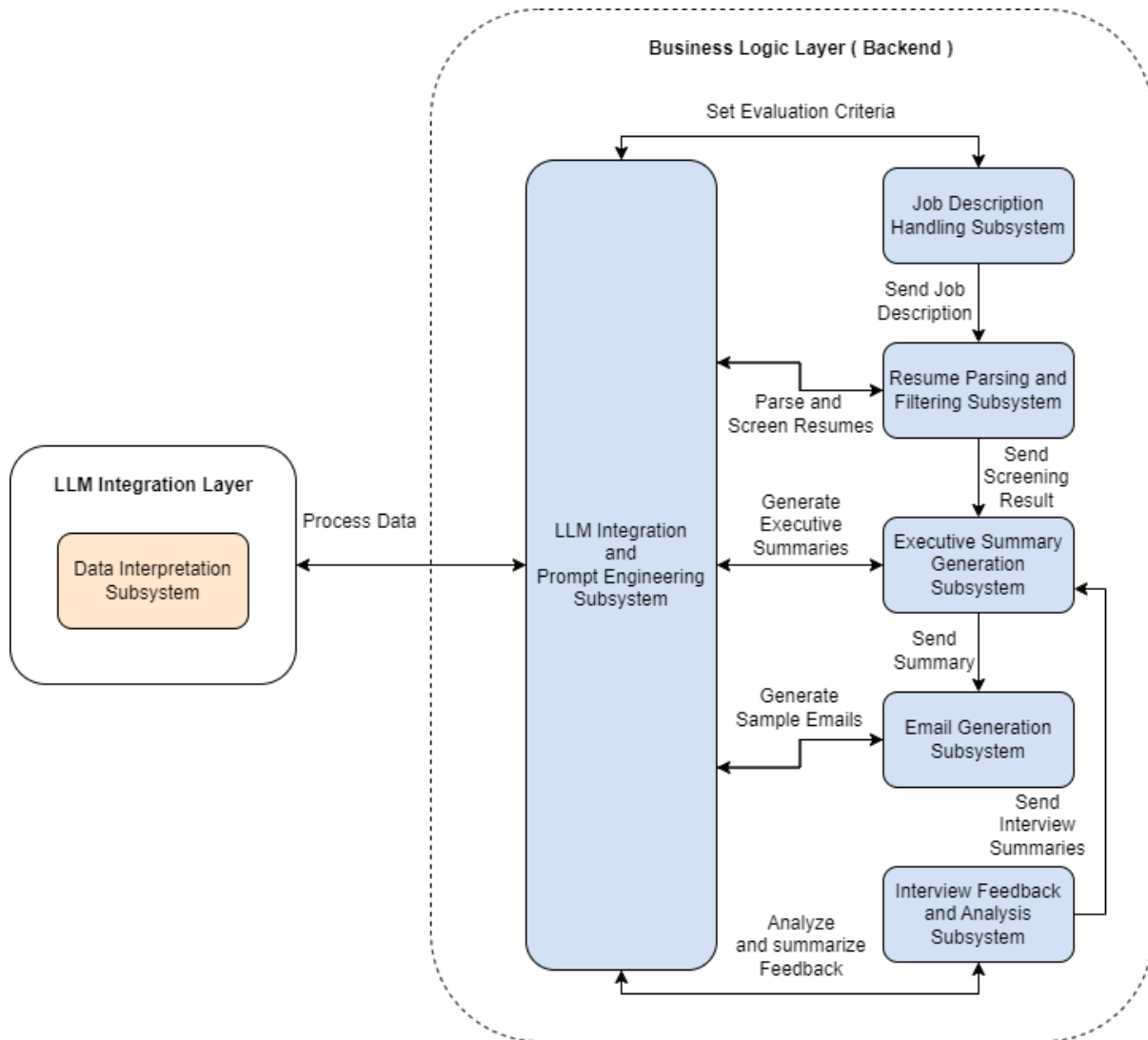


Figure 6.2: Back-end Services and LLM Integration Diagram

This diagram provides a detailed view of the back-end services' interaction with the LLM layer.

6.3.4.3 Detailed CRUD Operations

The diagram shows the CRUD operations that are performed by each subsystem, as well as the corresponding data entities in the database.

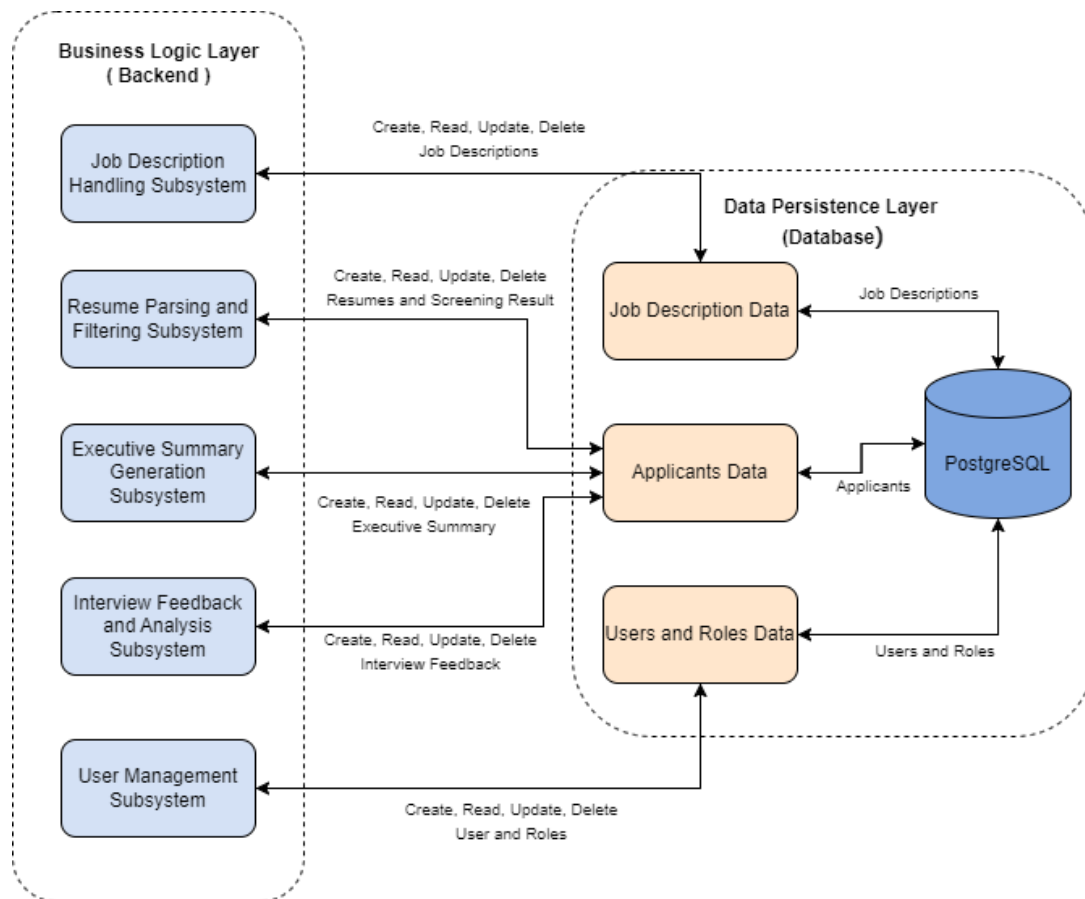


Figure 6.3: Interaction Between Back-end Services and Database Layer Diagram

This diagram shows the CRUD operations and interactions between back-end services and the database layer.

6.3.5 Subsystem Architecture

The following explains the details of the Low-Level Components in each Layer.

6.3.5.1 Presentation Layer Subsystems

The layer is divided into the following sub-components:

- **Authentication and Authorization Subsystem:** This subsystem is used to manage secure user access and permissions. Role-based access control is provided when users are correctly authenticated and their interactions with the application are by their authorization levels.
- **Dashboard Management Subsystem:** The primary interface of this subsystem is user-specific dashboards that constitute the different views for various user roles. It facilitates the effective monitoring and management of recruitment activities, providing a personalized experience for HR, Admins, and Interviewers.
- **Information/Data Handling Subsystem:** This manages the state of the application, handles error

resolution, and ensures the Front-End maintains consistent and up-to-date information.

6.3.5.2 Business Logic Layer Subsystems

The layer includes the following:

- **User Management Subsystem:** This subsystem is crucial for managing user accounts and is constructed using Django's user management framework.
- **Resume Parsing and Filtering Module:** This module uses the ability to use LLMs for structuring resume data. It facilitates efficient candidate screening by converting diverse resume formats into a JSON structure.
- **Job Description Handling Module:** It processes and analyzes job descriptions, also creates evaluation criteria and rules using LLMs.
- **Interview Feedback Analysis Module:** This module uses LLMs to provide an analytical overview of interview feedback, which can be used to produce comprehensive reports on candidates' performances.
- **Executive Summary Module:** This module is designed to generate concise and insightful summaries of candidate evaluations, and it integrates the strengths of LLM to assist in a quick and informed recap of the applicant process.
- **Email Generation Module:** Automates the process of creating and sending emails using a template engine and features for timely delivery.
- **LLM Integration Layer:** This layer would handle all the queries that need to be sent to LLM and get responses.

6.3.5.3 Database Subsystems

These subsystems are divided to focus on data storage and retrieval:

- **Data Storage Subsystem:** All permanent data stored by this subsystem e.g. user profiles, resumes, feedback from interviews, executive summaries, job descriptions, and other various data.
- **Data Retrieval and Query Processing Subsystem:** It is responsible for the efficient handling of queries coming from the back-end.

6.3.5.4 LLM Integration Subsystems

The subsystems interact with LLM models:

- **Data Interpretation Subsystem:** It is responsible for the processing of data obtained from the back-end, using Large Language Models (LLMs) to give outputs that we require.

6.4 Architectural Strategies

The section includes the strategies that affect the overall organization of the system and its higher-level structures.

6.4.1 Choice of Technologies

- **React JS for front-end:** ReactJS has been chosen for the front-end due to it being an efficient choice for dynamic and high-performing web application interfaces. Furthermore, it aligns well with the the goal of our application of creating a responsive and user-friendly interface.
- **Python & Django for back-end:** This framework have been selected for back-end development due to Python's versatility and Django's robustness. It facilitates rapid development and ease of integration with LangChain and the OpenAI API, thus proving to be a great back-end framework for applications harnessing the capabilities of LLMs and using prompt engineering.
- **PostgreSQL Database:** It is opted for its reliability and compatibility with complex data types, which was essential for managing extensive recruitment data.

6.4.2 Reuse of Existing Software Components

- **Tailwind CSS:** Employed for UI styling, enhancing the application's visual appeal and user experience.
- **CoreUI ReactJS:** The application UI shall be designed within the constraints of the CoreUI Design system because of its user-friendly design and smooth user interactions.

6.4.3 Future Plans for Extension or Enhancement

The architecture is designed to be modular and scalable, accommodating future enhancements and adapting to evolving recruitment practices. Software design principles will be employed in the making of the web application. Hence, the SOLID principles will be used. Especially the Open Closed principle will be used for allowing extension without modification.

6.4.4 Version Control

The application shall be hosted in the form of a collection of repositories on GitHub. The development team shall make pull requests and the changes will be incorporated upon approval of two other team members. This would improve the development process and make sure that the code goes through code reviews ensuring that the code is clean and follows the best industry practices.

6.4.5 User Interface Paradigms

A focus on responsive and intuitive UI employing ReactJS and Tailwind CSS supports the goal of accessibility and ease of use for a diverse user base. We will follow the Shneiderman's Eight Golden Rules of Interface Design that are as below:

- Strive for consistency
- Seek universal usability
- Offer informative feedback
- Design dialog to yield closure
- Prevent errors
- Permit easy reversal of actions
- Keep users in control
- Reduce short-term memory load

6.4.6 External Database and Data Storage Management

PostgreSQL's data storage and management capabilities support the system's growing data needs, ensuring data integrity and security.

6.4.7 Concurrency and Synchronization

On the front-end of the application async operations will be used to enable the user to interact with the website while the system works on other time-consuming tasks like web API requests. Moreover, at the backend, parallel processing will be used to ensure that the application is able to handle multiple user simultaneously. Django and Python's mechanisms ensure efficient handling of concurrent user interactions and system processes.

6.5 Domain Model/Class Diagram

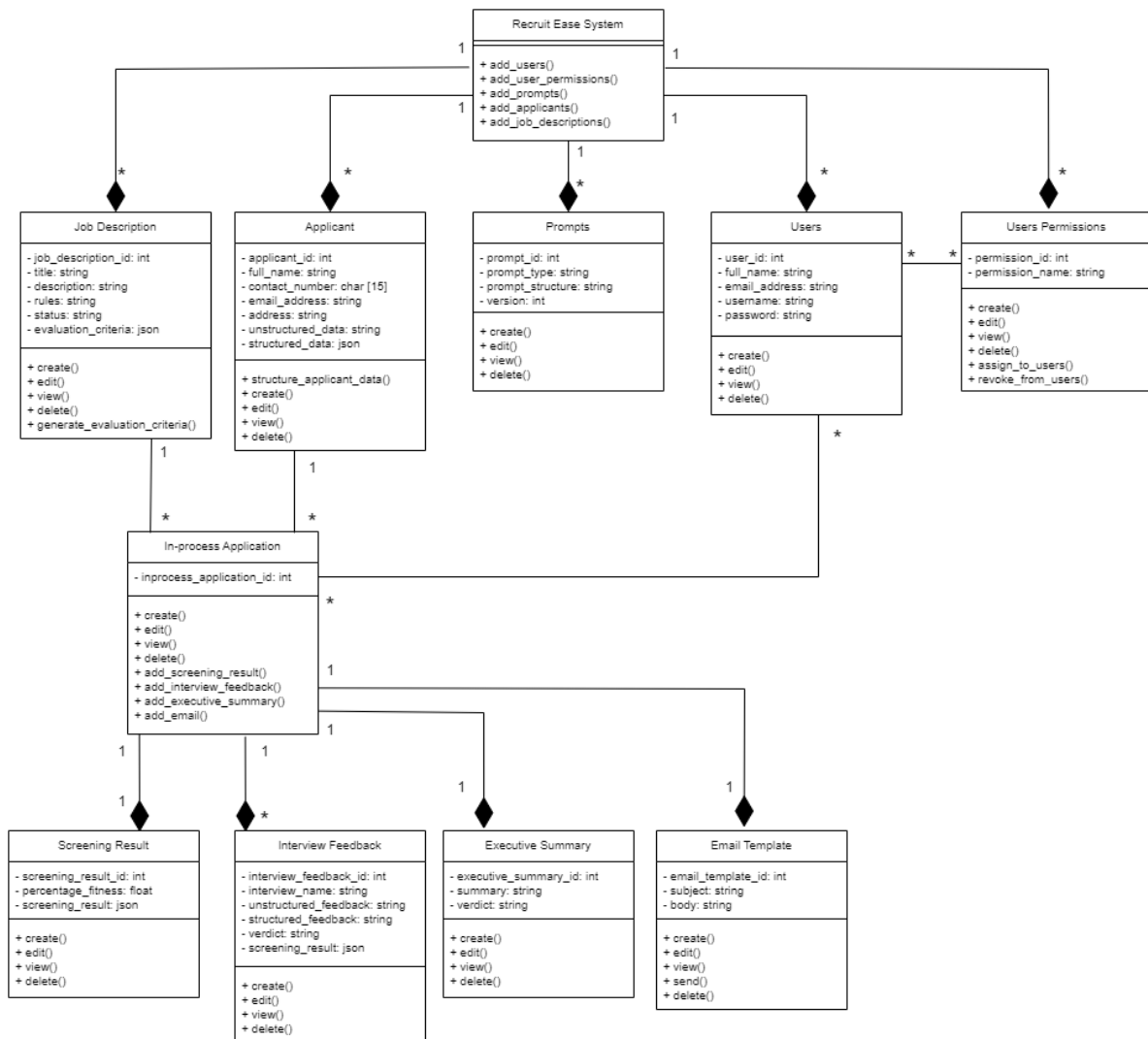
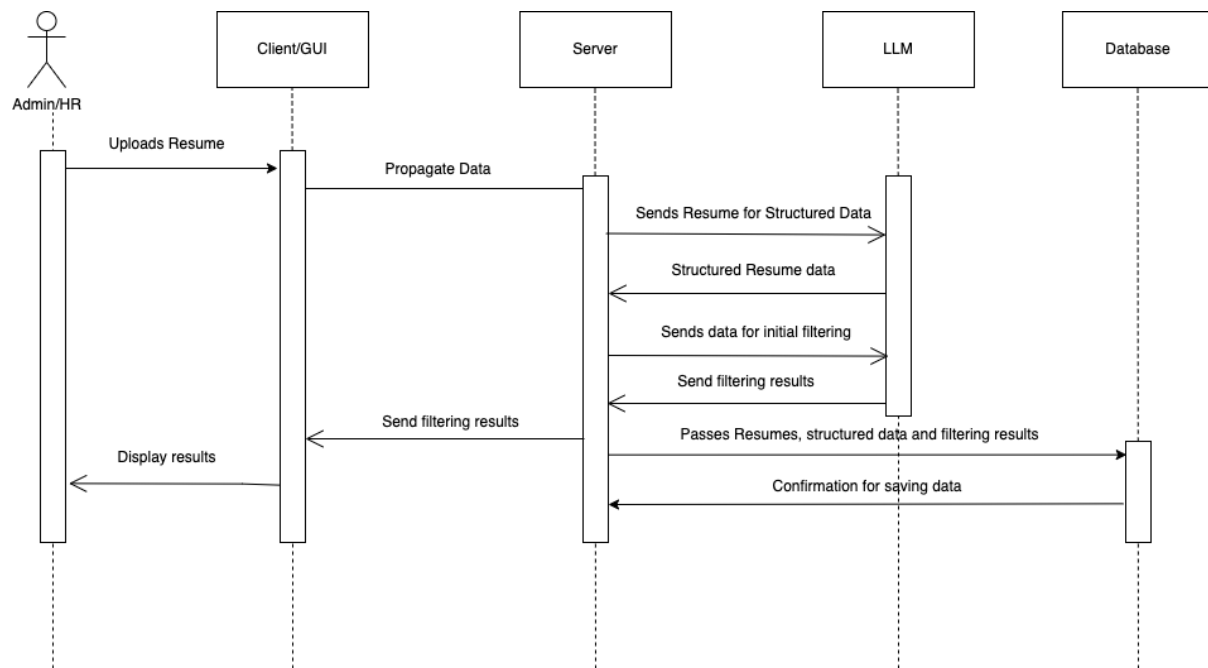
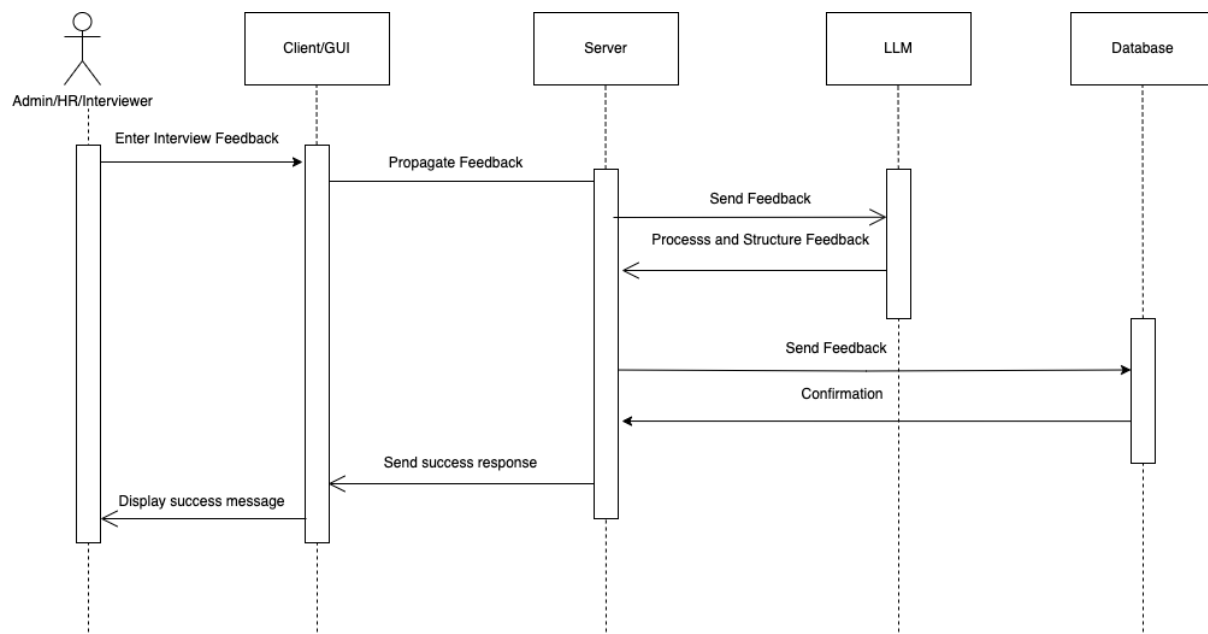


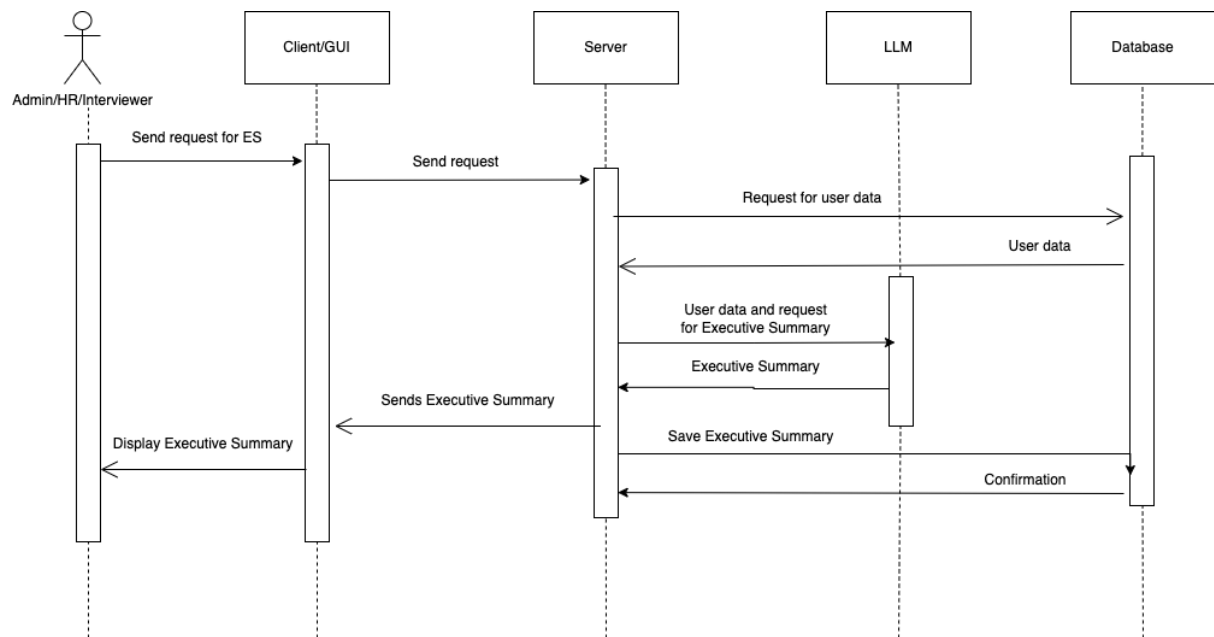
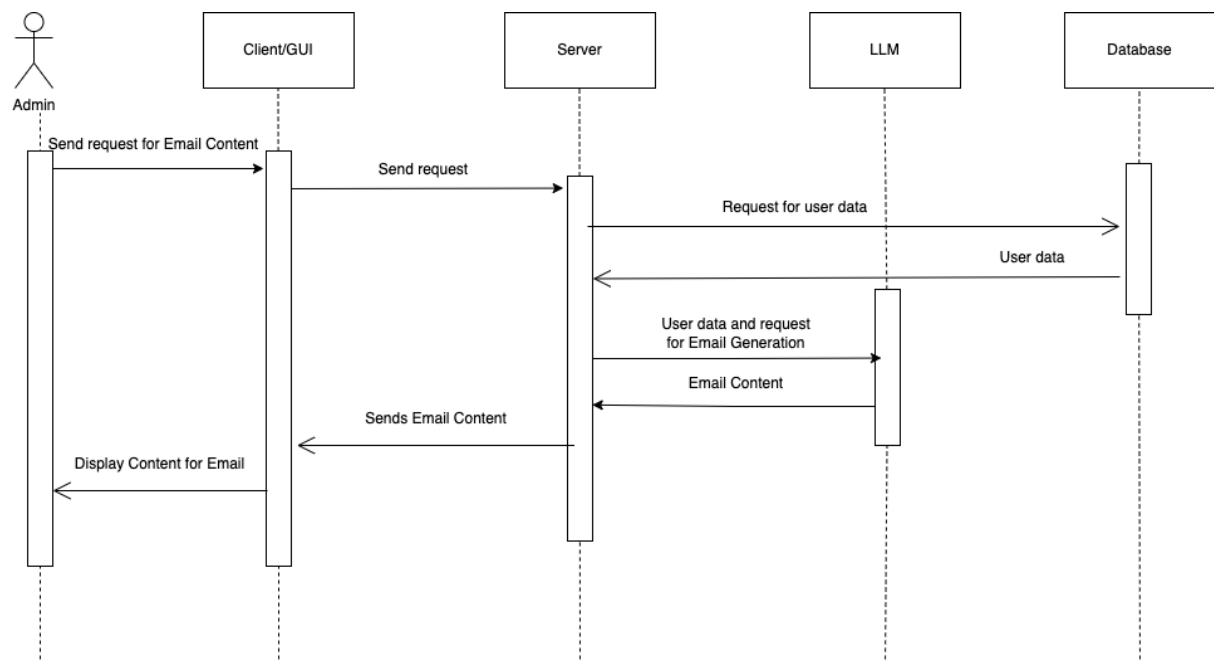
Figure 6.4: Class Diagram

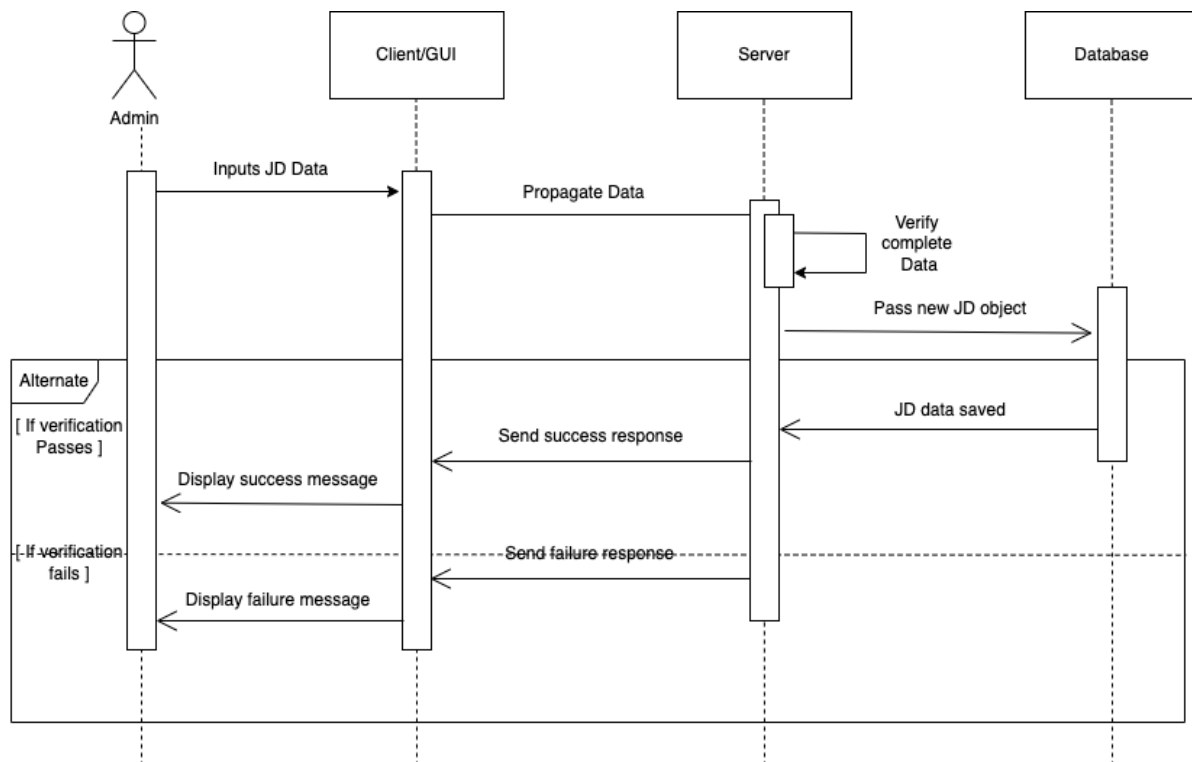
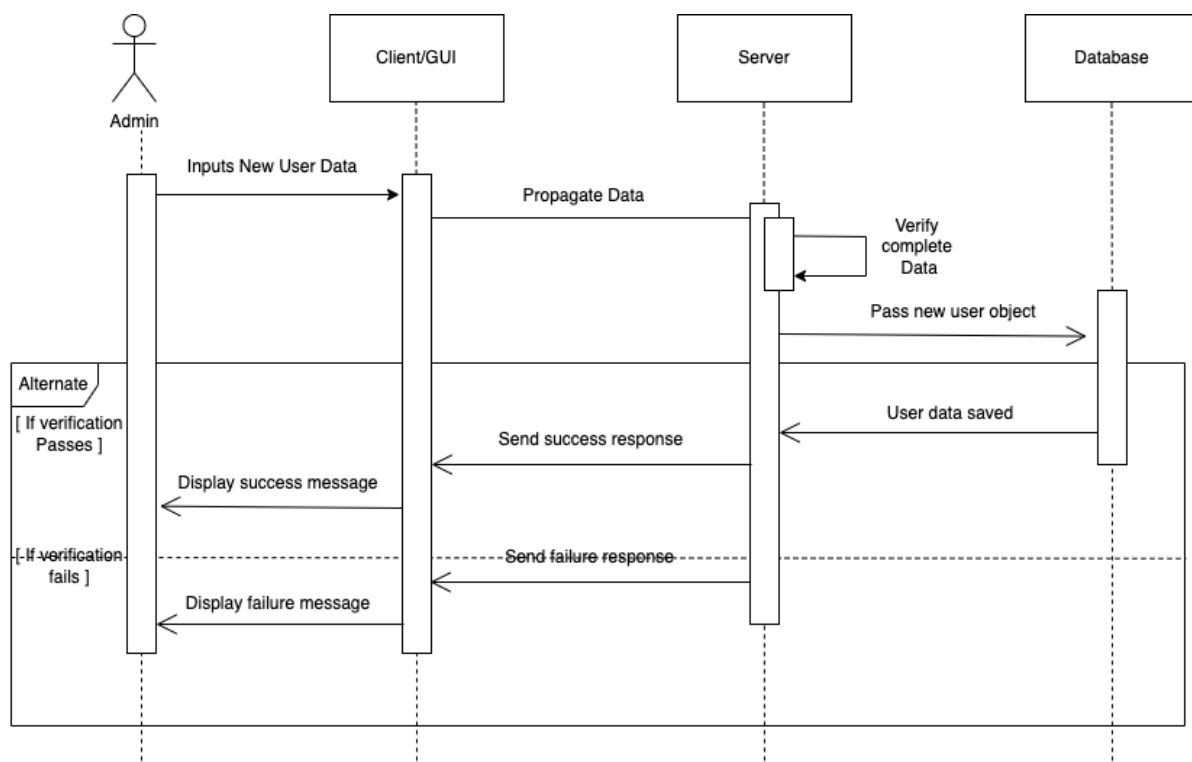
This figure represents all the entities and their relations

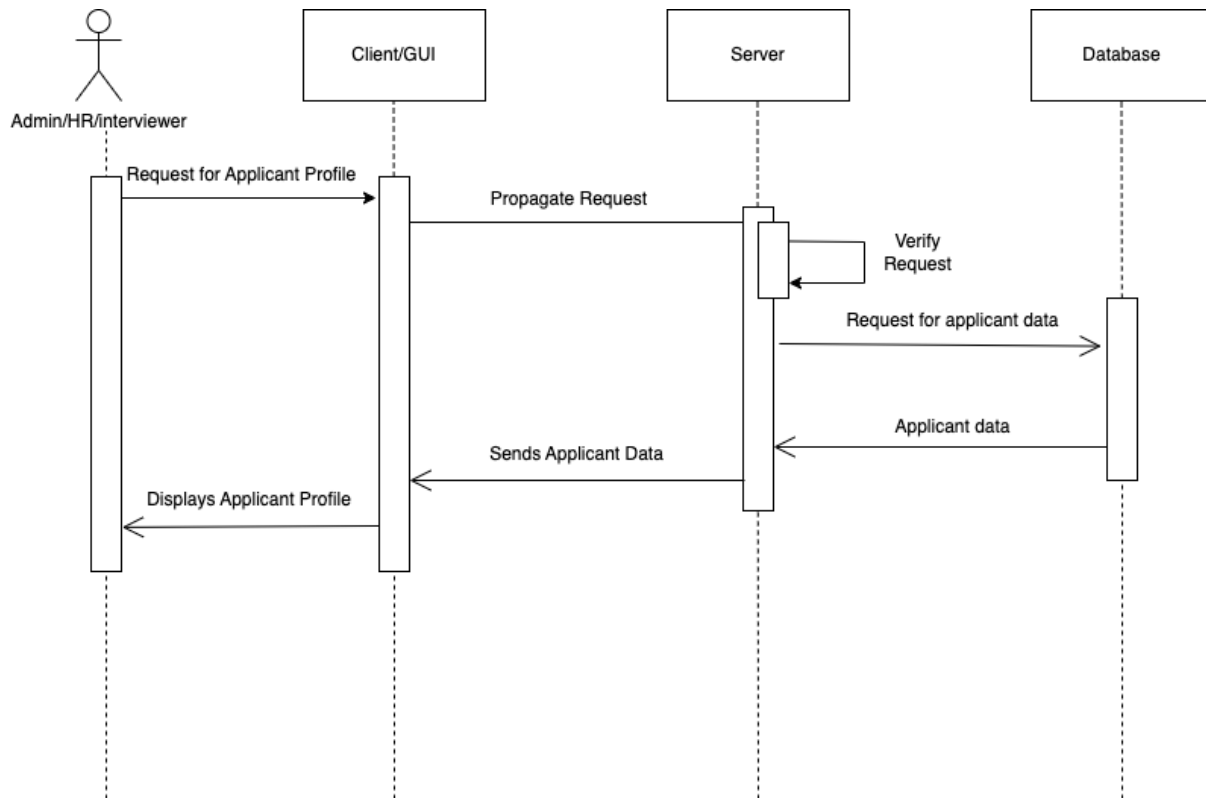
6.6 Sequence Diagrams

The following Sequence Diagrams will display the different objects in the system and how they exchange information between them.

**Figure 6.5: Upload Resume And Run Initial Filtering Diagram***HR wants to upload resumes and run initial filtering on them***Figure 6.6: Add Interview Feedback Diagram***Interviewer wants to add interview feedback*

**Figure 6.7: Generate Executive Summary Diagram***HR wants to generate executive summary***Figure 6.8: Generate Email Diagram***User wants to generate email content*

**Figure 6.9: Add New Job Description Diagram***User want to post a new job description***Figure 6.10: Add New User Diagram***Admin wants to create a new user*

**Figure 6.11: View Applicant Profile Diagram***User wants to view an individual applicant profile*

6.7 Policies and Tactics

To ensure the robustness and efficiency of the final application, the project employs various industry standards and policies. The following are some of the policies that we will follow during the software life cycle.

6.7.1 Coding Guidelines and Conventions

Since the project has been divided into two major portions: the back-end and the front end, the programming languages used for both will vary based on the requirements. The project adopts Python PEP 8 standards for the back-end and JavaScript ES6 standards for the front-end to ensure consistency, readability, and maintainability of the code. Industry Conventions and best practices are employed to ensure that the code written is well documented, quality and clean code. The IDE choice shall be VS Code due to the cross-compatibility and lightweight run time.

6.7.2 Testing Strategies

We will implement all types of testing which would include unit testing, integration testing, and end-to-end testing. This would ensure complete testing coverage of the application at different levels. We will use both white and black box testing techniques such as decision tables for black box and white box techniques such as UI, API and Unit testing.

6.7.3 Continuous Integration and Deployment

CI/CD pipelines are utilized for frequent code integration and automating testing and deployment, enhancing development cycle efficiency. GitHub will be used as the code repository with continuous additions of code to it and merges in the main branch whenever the code is reviewed to ensure that the new code is continuously integrated in the main stream.

6.7.4 Source Code Organization

The source code is hierarchically organized into modules and directories based on functionality, improving manageability and understanding of the code.

6.7.5 Error Handling and Recovery

Robust error handling and logging mechanisms are implemented to ensure system stability and quick recovery in case of failures.

6.7.6 User Interface Design

Responsive design principles using React JS and Tailwind CSS are adopted to ensure a seamless experience across different devices. The web application is made responsive for different screen sizes to ensure a smooth user experience and seamless experience for user irrespective of the device used to access the application.

6.7.7 Database Management

ORM in Django is used for database interactions, simplifying queries and reducing SQL injection vulnerabilities.

6.8 Conclusion

Recruit ease is a web based application designed to automate various stages of recruitment process. Application has been built while considering multiple design consideration to ensure ease of use and ease of learning. Furthermore, carefully designed architecture ensures modularity in the system. Various industry standards and policies have been used for further efficiency.

Chapter 7 Implementation and Test Cases

This chapter focuses on the development aspect of our application. We have integrated several major functionalities into the prototype each focusing on an important factor of the recruitment process.

7.1 Implementation

The implementation details of the components and algorithms designed in the application are given below.

7.1.1 Authentication

The Authentication component is an integral part of the system. This is the component that ensures secure access for users and prevents the application from unauthorized users. This system is to ensure that the user data remains safe and the integrity of the system remain intact. The front end for this is built using a React component for the Login screen, while the back end is in Django and the data base being PostgreSQL. The user credentials are stored in the data base in an encrypted form to ensure that no one can have unauthorized access to them.

7.1.2 Job Description Management

Since our system is an application to streamline and optimize the recruitment process, the management of the job positions and their respective job descriptions form an important component of the system. The front end for this component includes a couple of screens. The first one being the screen to add a new Job Positions. This screen displays detailed evaluation criteria with weightings assigned to individual sections once the job description is processed by the Large Language Model. The second one is the screen to view all job description available in the system's database and the third one is the screen to edit a JD that has already been saved in the system. The back end is responsible for storing and managing the data related to job descriptions. Overall, this component provides a smooth user experience and provides the users with a flexible way to interact with the application.

7.1.3 Resume Upload and Processing

The Resume Upload and Processing feature is an important aspect of the system. This component allows users to upload resumes against a job descriptions and then processes the request further. The component includes an initial screen where the user can upload a bunch of resumes against a particular job description and then forward them for further processing. Once the resumes are to be processed, the

data is extracted from them using external libraries and sent to the LLM with an additional prompt to structure them in relevant classes. Then there is a screen to display the results of this process. The result of this structuring is that no matter what the data of the resume is, it is presented in a clean and organized fashion ensuring that the data is easy to read and interpret.

7.1.4 Interview Forms and Feedback

The Interview Module is a vital component of the system, designed to enhance the recruitment process through the creation and management of dynamic interview forms. This module allows users to design customized forms using a drag-and-drop interface with various field options like text areas, radio buttons, and check boxes. Users can view, edit, or delete these forms through a dedicated management screen, ensuring the interview process is adaptable and up-to-date. Additionally, the module includes a recruitment steps template, which organizes multiple interview forms into sequences that align with the requirements of different job positions, and can be linked to one or more job descriptions. Furthermore, the interview feedback screen provides an interface to the interviewers where they can submit detailed feedback for each candidate, facilitating a structured evaluation process that aids in effective decision-making. Overall, the Interview Module streamlines the hiring workflow, making it more efficient and responsive to organizational needs.

7.1.5 Email Generation

The Email Generation Module is a crucial element in maintaining communication between applicants and HR personnel. This module begins with a screen where users can add email templates, incorporating placeholders for elements such as the job title and applicant name. These templates can range from generic communications such as acceptance or rejection emails to more customized interactions. There is also a functionality for editing or deleting existing templates, ensuring flexibility. Additionally, the module includes a screen for sending emails to applicants where users can choose to utilize a pre-existing template or compose a custom email, depending on specific needs. An advanced feature of this module is the ability to generate automated emails using Large Language Models, which can deliver detailed feedback directly to applicants. This feature not only streamlines the communication process but also ensures that feedback is comprehensive and delivered in a timely manner. Overall, the Email Generation Module enhances the efficiency and effectiveness of applicant-HR interactions, playing a vital role in the recruitment workflow.

7.1.6 Executive Summary Generation

The Executive Summary Generation module is a streamlined tool that automates the creation of concise and comprehensive summaries for each applicant, highlighting their key strengths and weaknesses. This module is crucial for facilitating quick and informed reviews of candidates by condensing the details of their recruitment process into an easily understandable format,

7.2 Test case Design and description

All the test cases performed to ensure proper functionality of our system during the testing phase are elaborated below.

Table 7.1: Login

This is the login process of our application

Login			
UC-01			
Test Case ID:	1	QA Test Engineer:	Maham Basit
Test case Version:	1	Reviewed By:	Fatimah Salman
Test Date:	28-04-2024	Use Case Reference(s):	4.6.1: Login
Revision History:	None		
Objective:	To check if user is successfully able to login to the system		
Product/Ver/Module:	Login module of website		
Environment:	Website is running on any web browser and internet is connected.		
Assumptions:	Login button is visible.		
Pre-Requisite:	The user is a valid user with credentials stored in the database.		
Step No.	Execution description	Procedure result	
1	User enters their credentials, i.e., their username and password, and clicks login.	System takes the user to the home screen.	
Comments: The test case is passed. Our system is working according to our need.			
Passed			

Table 7.2: Logout*This is the logout process of our application*

Logout			
UC-02			
Test Case ID:	2	QA Test Engineer:	Maham Basit
Test case Version:	1	Reviewed By:	Fatimah Salman
Test Date:	28-04-2024	Use Case Reference(s):	4.6.2: Logout
Revision History:	None		
Objective:	To check if user is successfully able to logout out of the system		
Product/Ver/Module:	Logout module of website		
Environment:	Website is running on any web browser and internet is connected.		
Assumptions:	Logout button is visible.		
Pre-Requisite:	The user is logged in the system.		
Step No.	Execution description	Procedure result	
1	User clicks the logout button	System takes the user to the login screen.	
Comments: The test case is passed. Our system is working according to our need.			
Passed			

Table 7.3: Sign Up New Users*This is the sign up process of our application*

Sign Up New User			
UC-03			
Test Case ID:	3	QA Test Engineer:	Maham Basit
Test case Version:	1	Reviewed By:	Fatimah Salman
Test Date:	28-04-2024	Use Case Reference(s):	4.6.3: Sign Up New Users
Revision History:	None		
Objective:	To check if admin is able to create a new users account		
Product/Ver/Module:	Signup module of website		
Environment:	Website is running on any web browser and internet is connected.		
Assumptions:	None		
Pre-Requisite:	The user is logged in the system and has the rights to create new user.		
Step No.	Execution description	Procedure result	
1	User clicks the create new user in the navigation bar.	System takes the user to create new user screen.	
2	User enters all the required details and clicks submit.	System creates new user in the back-end.	
Comments: The test case is passed. Our system is working according to our need.			
Passed			

Table 7.4: Add New Job Description

This process allows to add a new job description in our application

Add New Job Description			
UC-04			
Test Case ID:	7	QA Test Engineer:	Maham Basit
Test case Version:	1	Reviewed By:	Fatimah Salman
Test Date:	28-04-2024	Use Case Reference(s):	4.6.7: Add New Job Description
Revision History:	None		
Objective:	To check if user is able to add new job descriptions.		
Product/Ver/Module:	Job Description Module of website		
Environment:	Website is running on any web browser and internet is connected.		
Assumptions:	None		
Pre-Requisite:	The user is logged in the system and has the rights to create new job description.		
Step No.	Execution description	Procedure result	
1	User clicks the add new job description in the navigation bar.	System takes the user to add new job description screen.	
2	User enters all the required details and clicks submit.	System creates new job description in the back-end and shows the generated evaluation criteria.	
Comments: The test case is passed. Our system is working according to our need.			
Passed			

Table 7.5: Edit Job Description

This process allows to edit an existing job description in our application

Edit Job Description			
UC-08			
Test Case ID:	5	QA Test Engineer:	Maham Basit
Test case Version:	1	Reviewed By:	Fatimah Salman
Test Date:	28-04-2024	Use Case Reference(s):	4.6.8: Edit Job Description
Revision History:	None		
Objective:	To check if user is able to edit an existing job descriptions.		
Product/Ver/Module:	Job Description Module of website		
Environment:	Website is running on any web browser and internet is connected.		
Assumptions:	None		
Pre-Requisite:	The user is logged in the system and has the rights to edit job description.		
Step No.	Execution description	Procedure result	
1	User clicks the all job descriptions tab in the navigation bar	System takes the user to all job descriptions table screen.	
2	User clicks the edit icon in front of the job description he wants to edit.	System opens existing job description details in editable format.	
3	User edits the details and clicks submit.	System updates the details in the database.	
Comments: The test case is passed. Our system is working according to our need.			
Passed			

Table 7.6: Delete Job Description

This process allows to delete an existing job description in our application

Delete Job Description			
UC-09			
Test Case ID:	6	QA Test Engineer:	Maham Basit
Test case Version:	1	Reviewed By:	Fatimah Salman
Test Date:	28-04-2024	Use Case Reference(s):	4.6.9: Delete Job Description
Revision History:	None		
Objective:	To check if user is able to delete an existing job descriptions.		
Product/Ver/Module:	Job Description Module of website		
Environment:	Website is running on any web browser and internet is connected.		
Assumptions:	None		
Pre-Requisite:	The user is logged in the system and has the rights to delete job description.		
Step No.	Execution description	Procedure result	
1	User clicks the all job descriptions tab in the navigation bar.	System takes the user to all job descriptions table screen.	
2	User clicks the delete icon in front of the job description he wants to delete.	System deletes the job description from the database.	
Comments: The test case is passed. Our system is working according to our need.			
Passed			

Table 7.7: View All Job Descriptions

This process allows to view all existing job descriptions in our application

View All Job Descriptions			
UC-10			
Test Case ID:	7	QA Test Engineer:	Maham Basit
Test case Version:	1	Reviewed By:	Fatimah Salman
Test Date:	28-04-2024	Use Case Reference(s):	4.6.10: View All Job Descriptions
Revision History:	None		
Objective:	To check if user is able to view all existing job descriptions.		
Product/Ver/Module:	Job Description Module of website		
Environment:	Website is running on any web browser and internet is connected.		
Assumptions:	None		
Pre-Requisite:	The user is logged in the system and has the rights to view all job descriptions.		
Step No.	Execution description	Procedure result	
1	User clicks the all job descriptions tab in the navigation bar.	System takes the user to all job descriptions table screen.	
Comments: The test case is passed. Our system is working according to our need.			
Passed			

Table 7.8: Upload Resume

This process allows to upload resumes against job descriptions

Upload Resume			
UC-11			
Test Case ID:	8	QA Test Engineer:	Maham Basit
Test case Version:	1	Reviewed By:	Fatimah Salman
Test Date:	28-04-2024	Use Case Reference(s):	4.6.11: Upload Resume
Revision History:	None		
Objective:	To check if user is able to upload resumes against job descriptions.		
Product/Ver/Module:	Resume Module of website		
Environment:	Website is running on any web browser and internet is connected.		
Assumptions:	None		
Pre-Requisite:	The user is logged in the system and has the rights to upload resumes.		
Step No.	Execution description	Procedure result	
1	User clicks the upload resume tab in the navigation bar.	System takes the user to upload resume screen.	
2	User selects the job description, upload files and clicks submit.	System processes the resumes, stores them in the database and shows and takes user to the screening details screen.	
Comments: The test case is passed. Our system is working according to our need.			
Passed			

Table 7.9: Add Interview Feedback

This process allows to add Interview Feedback

Add Interview Feedback			
UC-13			
Test Case ID:	9	QA Test Engineer:	Mehak Farhan
Test case Version:	1	Reviewed By:	Fatimah Salman
Test Date:	28-04-2024	Use Case Reference(s):	4.6.13: Add Interview Feedback
Revision History:	None		
Objective:	To check if user is able to add interview feedback against an in-process applicant.		
Product/Ver/Module:	Interview Module of website		
Environment:	Website is running on any web browser and internet is connected.		
Assumptions:	None		
Pre-Requisite:	The user is logged in the system and has the rights to add interview feedback.		
Step No.	Execution description	Procedure result	
1	User navigates to the "Individual Applicant" section.	System displayed the "Individual Applicant" profile.	
2	User uploads the feedback document or type the text and clicks "Submit" button.	System generates summary and stored it in the database.	
Comments: The test case is passed. Our system is working according to our need.			
Passed			

Table 7.10: Generate Email*This process allows to generate emails for applicants*

Generate Email			
UC-14			
Test Case ID:	10	QA Test Engineer:	Mehak Farhan
Test case Version:	1	Reviewed By:	Fatimah Salman
Test Date:	28-04-2024	Use Case Reference(s):	4.6.14: Generate Email
Revision History:	None		
Objective:	To check if user is able to generate emails for in-process applicant.		
Product/Ver/Module:	Email Module of website		
Environment:	Website is running on any web browser and internet is connected.		
Assumptions:	None		
Pre-Requisite:	The user is logged in the system and has the rights to generate emails.		
Step No.	Execution description	Procedure result	
1	User navigates to the "Generate Email" section.	System displayed the "Generate Email" page.	
2	The User provides necessary inputs for the email.	System crafts the appropriate email based on the input or purpose.	
3	User clicks "Generate" button.	System presented the generated email to the User for review.	
Comments: The test case is passed. Our system is working according to our need.			
Passed			

Table 7.11: Generate Executive Summary

This process allows to generate executive summary of an in-process applicant

Generate Executive Summary			
UC-15			
Test Case ID:	11	QA Test Engineer:	Mehak Farhan
Test case Version:	1	Reviewed By:	Fatimah Salman
Test Date:	28-04-2024	Use Case Reference(s):	4.6.15: Generate Executive Summary
Revision History:	None		
Objective:	To check if user is able to generate executive summary of applicant.		
Product/Ver/Module:	Executive Summary Module of website		
Environment:	Website is running on any web browser and internet is connected.		
Assumptions:	None		
Pre-Requisite:	The user is logged in the system and has the rights to generate executive summary.		
Step No.	Execution description	Procedure result	
1	User navigates to the "Generate Executive Summary" section	System displayed the "Generate Executive Summary" page.	
2	The User selects the specific applicant in process.	System collated all relevant data and evaluations related to the selected applicant.	
3	User clicks "Generate" button.	System generated the executive summary and presented it to the User for review.	
Comments: The test case is passed. Our system is working according to our need.			
Passed			

Table 7.12: View All Applicants

This process allows to view all applicants

View All Applicants			
UC-16			
Test Case ID:	12	QA Test Engineer:	Mehak Farhan
Test case Version:	1	Reviewed By:	Fatimah Salman
Test Date:	28-04-2024	Use Case Reference(s):	4.6.16: View All Applicants
Revision History:	None		
Objective:	To check if user is able to view all applicants.		
Product/Ver/Module:	Applicant Module of website		
Environment:	Website is running on any web browser and internet is connected.		
Assumptions:	None		
Pre-Requisite:	The user is logged in the system and has the rights to view applicants.		
Step No.	Execution description	Procedure result	
1	The User navigates to the "View All Applicants" section	System retrieved all of the applicant data from the database and displayed it.	
Comments: The test case is passed. Our system is working according to our need.			
Passed			

Table 7.13: View an Individual Applicant

This process allows to view an individual applicant

View an Individual Applicant			
UC-17			
Test Case ID:	13	QA Test Engineer:	Mehak Farhan
Test case Version:	1	Reviewed By:	Fatimah Salman
Test Date:	28-04-2024	Use Case Reference(s):	4.6.17: View an Individual Applicant
Revision History:	None		
Objective:	To check if user is able to view an individual applicant.		
Product/Ver/Module:	Applicant Module of website		
Environment:	Website is running on any web browser and internet is connected.		
Assumptions:	None		
Pre-Requisite:	The user is logged in the system and has the rights to view applicants.		
Step No.	Execution description	Procedure result	
1	The User navigates to the "View All Applicants" section	System retrieved all of the applicant data from the database and displayed it.	
2	User selects the desired applicant.	System retrieved the detailed profile of the selected applicant from the database and displayed it.	
Comments: The test case is passed. Our system is working according to our need.			
Passed			

7.3 Test Metrics

The summary of the common ground of attributes of test case metrics is given below. 7.14.

Table 7.14: Functional Tests

This metric evaluates the performance of functional test cases

Metric	Purpose
Number of Test Cases	13.
Number of Test Cases Passed	13
Number of Test Cases Failed	0
Test Case Defect Density	0
Test Case Effectiveness	100
Traceability Matrix	File is enclosed separately.

7.4 Conclusion

The major functionalities provided by our application include authentication, job description management, resume uploading and processing, interview forms and feedback, and email generation. To ensure the proper functioning of all these components, we have designed several test cases. After running all

the test cases we ensured that our application will seamlessly meet the needs of all its users.

Chapter 8 User Manual

This chapter provides guidelines for the users of our application. This manual shall include instructions for all user roles.

8.1 User Roles and Permissions

- **Admin:** Has full access to all functionalities within the application.
- **HR:** Has access to most functionalities within the application, including managing Interviews, Job Descriptions, Applications, User Management, and Emails.
- **Interviewer:** Has limited access and can only review and provide feedback on applicants assigned to them.

8.2 General User Operations

This section provides guidelines for common functionalities for all users, such as login, logout and basic navigation tips.

8.2.1 Login

To access the application, follow the following steps:

- Open your web browser and navigate to the login page of RecruitEase
- Enter the email and password provided by the Admin or HR.
- Click on the Login Button.

8.2.2 Logout

To end your session, follow the following steps:

- Click on the User Icon on the top right of the navigation bar.
- Click the 'Logout' option from the drop-down.

8.2.3 Basic Navigation Tips

The layout of the application is simple to navigate and access. It has the following basic components:

- **Dashboard:** The user is directed to the dashboard upon logging in, which provides custom key metrics and stats according to the user and their role.
- **Navigation Bar:** This is located on the top of the application and includes basic buttons to navigate to the dashboard, settings, and user profile. It can open the additional sidebar through the burger icon.
- **Side Bar:** This is located on the left of the application and includes all the different modules different roles can access. It can be accessed by clicking the burger icon on the navigation bar.

8.3 HR User Manual

An HR has access to all modules of the Application. HR has to manage the Interviews, Job Descriptions, Applications, User Management, and Emails.

8.3.1 Interview Module

This section provides step-by-step guidelines for configuring the Interview Module, including creating interview types and recruitment steps templates.

8.3.1.1 Add Interview Type

Define various types of interviews by customizing forms and structures dynamically to suit specific job roles and organizational needs.

- Navigate to the sidebar menu and select "Interview Module."
- Click on "Add Interview Type".
- Enter Interview Type name
- Utilize the dynamic drag-and-drop feature to design the interview form according to your requirements.
- Drag different form options to create the desired structure and content for the interview form.
- Click the 'Submit' button to finish adding the interview type.

8.3.1.2 Add Recruitment Steps Template

Create a standardized template for the recruitment process by defining sequential steps, duration, and associated interview types.

- Navigate to the sidebar menu and select "Interview Module."

- Click on "Add Recruitment Steps Template".
- Enter Template name
- Click on "Add Step" for each interview to be conducted within the template.
- For each step, enter the name, duration, and select the corresponding interview type created earlier.
- If you need to remove any steps, click the delete button next to each step.
- Click the 'Submit' button to finish adding the recruitment steps template.

8.3.2 Job Description Module

This section provides step-by-step guidelines for creating job descriptions within the Job Description Module.

8.3.2.1 Add Job Description

Create and add Job Descriptions and generate evaluation criteria for each.

- Navigate to the sidebar menu and select "Job Description."
- Click on "Add Job Description."
- Enter the Job Description name.
- Provide a comprehensive description of the job role.
- Specify any specific rules or requirements using a bulleted list format to address biases that cannot be included in the description.
- Select the recruitment steps template to be followed from the drop-down menu.
- Choose the status of the job description (open or closed).
- Click the "Submit" button to finalize the job description.
- Wait for the evaluation criteria to be generated based on the selected template.

8.3.3 Application Module

This section provides step-by-step guidelines for using the Application Module, including uploading resumes, reviewing screening results, and managing in-process applicants.

8.3.3.1 Upload Resume

Upload resumes for job applications and wait for each resume to be evaluated.

- Click on "Upload Resumes" from the sidebar menu.
- Select the Job Description from the dropdown menu.
- Drag and drop or select multiple resumes in PDF or .docx format.
- Click "Submit."
- Wait for each resume to be evaluated.

8.3.3.2 Screening Results

Review screening results for job applications and shortlist applicants for further processing.

- Click on "Screening Results" from the sidebar menu.
- Select the Job Description card for which you want to see the screened applications.
- In the table, view the applicant's name, total score, and sub-scores.
- Click on an applicant in the table to see a detailed view of the screening results in a pop-up window.
- In the pop-up window, view the applicant's resume on the left and two accordion options on the right: "Evaluation Criteria" and "Rules."
- Click on "Evaluation Criteria" to see how the application assigned scores according to the Job Description. The evaluation criteria will show scores and sub-criteria that the applicant has passed or failed.
- Click on "Rules" to see how the application passed each rule or not. The green tick indicates that the applicant passed the evaluation criteria and the rule.
- Click the "Accept" or "Reject" button from the pop-up to shortlist the applicant for further processing.

8.3.3.3 In-process Applications

Manage and review in-process applications within the Application Module.

- Click on "Applications" from the sidebar menu.
- Select "All Applications" to view all applications.
- Choose the Job Description card to see shortlisted applicants.
- The table displays the applicant's name, email address, status, and an option to view details.

- Click "View Details" to see the application in detail.
 - The applicant's Resume is shown on the left.
 - Analyze the progress chart to track the applicant's journey through the recruitment process.
 - On the right side, you'll find accordions for detailed information.
 - In the "Screening Results" accordion, view the results of the applicant's screening process.
 - In the "Interview Types" accordion (provided by the recruitment steps template), you have the following options to click "Add Interview Feedback" to provide feedback for the interview, click "View Interview Feedback" to see feedback provided by the interviewer and click "Generate Summary" to generate a summary of the interview feedback.
 - Click "Schedule Interview" to set up an interview schedule. Select collaborators for the interview from the drop-down menu and add the scheduled date and time for the interview.
- To generate an executive summary for the applicant's entire profile and progress, click on the "Generate Executive Summary" button.
- To generate an email for the applicant, click on the "Generate Email" button. This action navigates to the email generation module.

8.3.4 Executive Summary Module

This section provides step-by-step guidelines for generating executive summaries for applicants.

8.3.4.1 Generate Executive Summary

Generate a comprehensive executive summary for an applicant.

- Click on "Applications" from the sidebar menu.
- Select "All Applicants".
- Choose the applicant for whom you want to generate the executive summary.
- Click "View Details" for the applicant.
- Click on the "Generate Executive Summary" button.
- A pop-up window will appear displaying the executive summary.

8.3.5 User Management Module

This section provides step-by-step guidelines for managing users.

8.3.5.1 Register User

Register new users within the system.

- Click on "User Management" from the sidebar menu.
- Click on "Register User."
- Enter the User name, First and Last name, Email, Password, and re-enter the password.
- Select the role from the drop down menu.
- Click "Register" to create the user account.

8.3.6 Email Module

This section provides step-by-step guidelines for managing email templates within the Email Module.

8.3.6.1 Add Email Template

Create and customize email templates for efficient communication with applicants.

- Click on "Email Templates" from the sidebar menu.
- Click on "Add Email Template."
- Enter the email template name.
- Optionally, specify email addresses for CC or BCC.
- Enter the subject line of the email.
- Compose the body of the email using the provided text editor with styling options.
- Add placeholders within the body text to personalize emails, such as {applicant_name} or {job_description}.
- Click on "Add" to save the email template.

8.3.6.2 Generate Email for Applicants

Send personalized emails to applicants using predefined email templates.

- Click on "All Applicants" from the sidebar menu.
- Select the Job Description for the applicants you want to send emails to.
- Click "View Details" for the applicant you want to generate an email for.

- Click on the "Generate Email" button.
- Select the desired email template from the drop-down menu.
- Customize the pre-filled template as necessary.
- Click "Send Email" to send the email to the applicant or click "Save Draft" to save it as a draft.
- To view saved drafts, click on "View Saved Drafts."

8.4 Interviewer User Manual

An Interviewer has access to review and provide feedback on applicants assigned to them.

8.4.1 Reviewing and Providing Feedback

This section provides step-by-step guidelines for reviewing applicants and adding interview feedback.

8.4.1.1 Add Interview Feedback

Provide feedback on applicants while conducting interviews.

- Click on "Applicants" from the sidebar menu.
- Click on "All Applicants."
- Select the Job Description card to view applicants assigned for interviews.
- Click on "View Details" for the applicant you are interviewing with.
- Click on the accordion for the interview to expand it.
- Click on "Add Interview Feedback."
- The applicant's resume is displayed on the left for reference.
- Fill out the feedback form according to the interview performance.
- Click on "Add Feedback" to finalize and submit the feedback.

Chapter 9 Experimental Results and Discussion

In this section we have mentioned multiple experiments we carried out on Large Language Models to generate accurate results. Unstructured resume data used is given below.

```
applicant_resume = """
Maham Basit
Computer Science Senior mahambasit01@gmail.com 03354388894 Lahore, Pakistan linkedin.com/in/maham-basit
An ambitious and passionate CS Senior looking to work in a progressive environment, to learn and explore App and Web develop
EDUCATION
BS Computer Science FAST NUCES Lahore Campus
2020 - Present, 3.99 CGPA
GCSE Aleveles Beaconhouse College Program Johar Town
2018 - 2020, 4A*
EXPERIENCE
Software Engineering Intern Conrad Labs
2023-2023, Teaching Assistant
FAST NUCES BSCS
2022 - 2023, Data Structures and Data Bases Competition Coordinator
SOFTEC
2023 - Present, Biggest IT event held at FAST NU Lahore Volunteer Team Lead
Future Fest
2023, Biggest IT conference event in Pakistan Assistant Vice President
STACKS FAST Chapter
2022 - 2023
CERTIFICATES
Technical Writing Workshop (2023)
Organized by Educative
Introduction to Cloud Computing (2022 - 2023)
AWS Course MERN Bootcamp (2022)
Organized by TinTash
SKILLS
C++ OOP Data Structures SQL Java Spring Boot Asp.Net HTML CSS
Assembly Language Solidity Python MERN Problem Solving Communication Teamwork
PERSONAL PROJECTS
Textile Mill Salary Management System
Project with database concepts of SQL, backend in C# and frontend in HTML CSS on Asp.net framework.
Brick Breaker
Console based game in C++ designed using concepts of Programming fundamentals and basic graphics.
Ultimate Tic Tac Toe
Console based game in C++ designed using concepts of Object Oriented Programming.
Game of Life Game designed in C++ using concepts of Object Oriented Programming and SFML (Super Fast Multimedia Library) for
Transport Management System
Project in C++ using concepts of Object Oriented Programming.
Character Crush
Game similar to candy crush designed in Assembled Language.
ACHIEVEMENTS
Deans List of Honor's Fall 2020 & Spring 2021
Rectors List of Honor's Fall 2021, Spring 2022, Fall 2022 & Spring 2023
INTERESTS
Web Development App Development MERN Stack
"""
```

Figure 9.1: Unstructured Resume 1

This is the unstructured data extracted from a resume for experiments

```

applicant_resume = """
Fatimah Salman
STUDENT . BS COMPUTER SCIENCE . FAST NU LAHORE
Address: Lahore, Punjab Email: fatimahsalman776@gmail.com
LinkedIn: https://www.linkedin.com/in/fatima-salman-1836b11b5/ Phone No: 0335-4477366
About Me
I'm a BS Computer Science Undergraduate student at FAST NUCES. I'm a proactive and smart working learner. I love to take up
challenge and solve complex problems. I do not restrict myself to known technologies rather I love to explore and grasp new
Education
FAST NUCES Lahore BS(Computer Science) Aug 2020-July 2024(expected)
CGPA: 3.99
• Currently in my 6th semester. Have studies core programming subjects like Fundamentals, OOP, Data
Structures, Operating Systems, Algorithms, Database Systems Software Design and Analysis
Punjab College
Intermediate Sep 2018-June 2020
Marks: 991/1100
Lahore Grammar School
Matriculation Jan 2016-July 2018
Marks: 1056/100
Skills
Programming: C++ . Python . JavaScript(basics) . SQL . C
Web/App: React. HTML . CSS
Tools: Visual Studio . Atom . MS Word . Powerpoint .Google
Additional Courses
• The Complete REACT with Redux by Stephan Grider, Udemy July 2022 - Sep2022
50+ hours which taught basic web development concepts (Html, Css, JavaScript) to advance
React and Javascript functionalities
• AWS Course of Cloud Computing
A course that covers all the basics of cloud computing and AWS basics
Experience
Teaching Assistant - Spring 2023 Operating Systems- Mr. Saifullah Tanvir
Led discussions and review sessions, graded assignments and quizzes, provided individual support to students and assisted
with course planning as Teaching Assistant
Projects
1. Basic Banking System
2. Employee Management System - SQL and C#
3. Portfolio Website
4. A complete functional terminal in C
Achievements
Rector's list of Honors Dean's list of Honor
Fall 2020, Fall 2021, Spring 2022, Fall 2022 Spring 2021 SGPA:3.96
Extra Curricular Activities
• Manager at Career Counseling Society
Organized events like Annual Job Fair, NUCES Talks, Seminars, Webinar, recruitment drives etc
• Exceptional Volunteer SOFTEC
Exceptional Volunteer for SOFTEC programming Competition 2022
• Host for Official events of university
Hosted official events of the university like Convocation and Prize Distribution Ceremonies"""

```

Figure 9.2: Unstructured Resume 2

This is the unstructured data extracted from a resume for experiments

9.1 Initial Experiment

The initial experiment was designed as a proof of concept to assess the Large Language Model's capabilities in the automated recruitment process. Our aim was to understand how effectively the LLM could perform the role of HR specialist. In this experiment, we used a role-based technique. This technique gives purpose and direction to LLM so it can make more focused responses. We assigned the LLM the role of an HR specialist to contextualize its task. We provided LLM with an applicant's resume and four criteria points. Criteria points were clearly stated to minimize any ambiguity. One of the criteria regarding candidates work experience was slightly complex. The criteria was not just about having work experience but also specified minimum years to be 5. The task for the LLM was explicit: to accept the candidate if all the given criteria were met, or to reject them otherwise. The results highlighted a significant limitation in the LLMs current processing capabilities. LLM identified the presence of work experience in the resume, but failed to see the minimum experience required. Resume had experience of only one year but LLM did not consider it. This showed that LLM appeared to selectively process

parts of the criteria. Hence, LLM was currently unable to interpret complex rules leading to incorrect or incomplete evaluations of the candidates.

```
resume_prompt = """
You are a talent acquisition HR and your task is to Accept/Reject the following resume = {resume}.

Requirements are as follows:

- GPA greater and equals to 3.5
- Should be from a renowned university like FAST, NUCES, NUST, LUMS, GIKI, PUCIT
- Must have past work experience only relating to development of 5 or greater than 5 years
- Must have created a project for web development

Accept the candidate if all requirements are met, else reject.
"""
```

Figure 9.3: First Experiment Prompt

This is the first prompt we used to evaluate resume against four criteria points

```
Based on the given requirements, I would accept the candidate Maham Basit. She meets the following criteria:

- GPA of 3.99 which is greater than 3.5
- Studying at FAST NUCES Lahore Campus which is a renowned university
- Has relevant work experience as a Software Engineering Intern at Conrad Labs
- Has created a personal project for web development (Textile Mill Salary Management System)
```

Figure 9.4: First Experiment Result

This is the result we received from LLM against the first prompt

9.2 Implementing Chain of Thought Prompting

Considering the limitations observed in the first experiment we implemented Chain of Thought Prompting. Chain of Thought prompting involves breaking down a large complex problem into smaller sub problems each of which is then considered separately by the LLM. We structured our prompt to ask LLM to evaluate each criterion individually, providing a clear 'Yes' or 'No' response along with its reasoning for each decision. We provided LLM with the same resume and criteria. This step by step approach helped LLM to process each information with greater attention and precision. It first gave Yes and No with reason for each criteria. Then it compiled all the individual decisions to make the final decision. By following this structured approach it gave the correct response.


```

resume_prompt = """
You are a talent acquisition HR and your task is to Accept/Reject the following resume = {resume}.

Step 1: See that it fits all of the following criteria:

- GPA greater and equals to 3.5
- Should be from a renowned university like FAST, Nukes, Nust, Lums, GIKI, PUCIT
- Must have past work experience relating to development of 5 or greater than 5 years
- Must have created a project for web development

Step 2: For every criteria evaluate the candidate. The answer should be in the format:

- YES/NO
- Reason: short and valid reason in bullet points with related to standards set by step 1

Step 3:
- Accept/Reject
- Accept if candidate has all Yes, Reject even if there is 1 No

The resume starts from here:

"""

```

Figure 9.5: Chain of Thought Prompt

This is the prompt with CoT prompting introduced to evaluate resume against four criteria points

```

Step 1:
- GPA greater and equals to 3.5: YES
- Should be from a renowned university like FAST, Nukes, Nust, Lums, GIKI, PUCIT: YES
- Must have past work experience relating to development of 5 or greater than 5 years: NO
- Must have created a project for web development: YES

Step 2:
- GPA greater and equals to 3.5: YES
  - Reason: The candidate has a 3.99 CGPA which is greater than the required 3.5 GPA.
- Should be from a renowned university like FAST, Nukes, Nust, Lums, GIKI, PUCIT: YES
  - Reason: The candidate is from FAST NUCES Lahore Campus which is a renowned university.
- Must have past work experience relating to development of 5 or greater than 5 years: NO
  - Reason: The candidate has only worked as a Software Engineering Intern and Teaching Assistant for a short period of time.
- Must have created a project for web development: YES
  - Reason: The candidate has created a Textile Mill Salary Management System project using SQL, C#, HTML, and CSS on Asp.net framework.

Step 3:
- Reject: The candidate does not meet the criteria of having past work experience relating to development of 5 or greater than 5 years.

```

Figure 9.6: Chain of Thought Result

This is the result we received from LLM against the CoT prompt

9.3 Testing Real-World Scenarios

In our third experiment we tested the prompt with a real-world job description. Real-world job descriptions often contain complex details and requirements, posing a significant challenge for the LLM. The experiment was to evaluate two basic requirements. Firstly, to see how effectively the LLM could extract, interpret, and match these complex job descriptions against candidate profiles. Secondly, to test the consistency of the LLM by giving the same job description and resume to it multiple times. Results showed that LLM was unable to completely check all requirements from detailed and complex job descriptions. Furthermore, despite the inputs being the same, the LLM produced varying results. This highlighted a significant challenge in ensuring a consistent and fair screening process. To address these challenges we decided to break our problem into three parts.

- **Structured Resume:** This was to structure resumes into categories such as skills, education, experience, etc.

- **Job Description Criteria:** This was to structure job description and create an evaluation criteria. This would also include categories like skills, education, experience etc.
- **Screening:** Matching structured resumes with structured job descriptions. The structured format of both will help in easier and more accurate comparison by standardizing the format.

```

resume_prompt = """
You are a talent acquisition HR and your task is to rate the following resume = {resume}.

Step 1: See that it fits all of the following job description = {job_description}:

Step 2: For every job description point evaluate the candidate. The answer should be in the format:

- YES/NO
- Reason: short and valid reason in bullet points with related to standards set by step 1

Step 3:
- Accept/Reject
- Accept if candidate has all Yes, Reject even if there is 1 No

The resume starts from here:
"""

```

Figure 9.7: Real World Job Description Prompt

This is the prompt we used to evaluate resume against an actual job description

```

Step 1: The job description requires a candidate with a BS/MS in Computer Science or related field, 2+ years of industry experience in Software Engineer role, good problem-solving skills, strong hands-on experience with Java, Python or Ruby, experience building RESTful APIs, hands-on web-development experience, strong DB design and analysis skills, development experience with Tomcat, MySQL or similar tools, a good command on source control (Git, SVN) and understanding of defect tracking systems, experience with ORM frameworks like JPA, Hibernate, EJB, experience with coding test cases, good communication skills working with small, cross-functional, and remote teams, and a genuine excitement for software development.

Step 2:
- YES: Maham has a BS in Computer Science and a high CGPA.
- YES: Maham has experience as a Software Engineering Intern and as a Teaching Assistant.
- YES: Maham has experience with C++, Java, Python, and SQL.
- NO: Maham does not have specific experience with Spring, Django/Flask, or Rails.
- YES: Maham has experience with HTML, CSS, and Javascript.
- YES: Maham has experience with DB design and analysis skills.
- NO: Maham does not have specific experience with Tomcat or MySQL.
- YES: Maham has experience with Git and SVN.
- NO: Maham does not have specific experience with ORM frameworks like JPA, Hibernate, EJB.
- YES: Maham has experience with coding personal projects.
- YES: Maham has good communication skills and has worked in volunteer and team lead positions.
- YES: Maham has expressed a genuine interest in software development.

Step 3: Accept. Maham has all the required qualifications and experience, except for specific experience with certain tools and frameworks. However, she has demonstrated the ability to learn and work on personal projects, which shows her potential to adapt to new technologies.

```

Figure 9.8: Real World Job Description Result

This is the result we received from LLM against actual job description

9.4 Structuring Resumes

This experiment focused on determining the capability of LLM to organize and structure resumes into categories. A set of common categories such as education, skills, experience was provided to LLM. However, it was not restricted to using only these categories, allowing some degree of flexibility in its approach. This showed major drawbacks. The LLMs frequently deviated from the provided categories, sometimes creating new ones and repeating existing ones. This led to a lack of consistency across different resumes. Each resume processed by the LLM ended up with a different set of categories. This was

due to multiple reasons including different categories in different resumes and lack of strict guidelines to the LLM. Furthermore, data from the resume was not properly categorized under relevant headings. Skills, experience and projects were mixed and points were repeated under these. This created confusion and inaccuracy. The experiment highlighted the need for more precise guidelines to avoid variations.

```
resume_prompt = """
Interpret and structure the content from text={resume}.
Resume can be categorized into different headings like:

1. Contact Information
2. Education
3. Work Experience
4. Extra-curricular Activities
5. Projects
6. Skills
7. Certifications
8. Honors and Awards
9. Languages
"""
```

Figure 9.9: Structuring Resume Prompt

This is the prompt we used to structure the resume into categories

```
Based on the content provided, the resume can be structured as follows:

Maham Basit
Computer Science Senior
Email: mahambasit01@gmail.com
Phone: 03354388894
Location: Lahore, Pakistan
LinkedIn: linkedin.com/in/maham-basit

Objective:
An ambitious and passionate CS Senior looking to work in a progressive environment, to learn and explore App and Web development. Eager to gain industry experience through internships before graduating.

Education:
- BS Computer Science, FAST NUCES Lahore Campus, 2020 - Present, 3.99 CGPA
- GCSE Alevels, Beaconhouse College Program Johar Town, 2018 - 2020, 4A*

Work Experience:
- Software Engineering Intern, Conrad Labs, 2023
- Teaching Assistant, FAST NUCES BSCS, 2022 - 2023
- Data Structures and Data Bases Competition Coordinator, SOFTEC, 2023 - Present
- Volunteer Team Lead, Future Fest, 2023
- Assistant Vice President, STACKS FAST Chapter, 2022 - 2023

Certifications:
- Technical Writing Workshop (2023), Organized by Educative
- Introduction to Cloud Computing (2022 - 2023), AWS Course
- MERN Bootcamp (2022), Organized by TinTash

Skills:
- C++, OOP, Data Structures, SQL
- Java, Spring Boot, Asp.Net, HTML, CSS
- Assembly Language, Solidity, Python
- MERN, Problem Solving, Communication, Teamwork

Projects:
- Textile Mill Salary Management System
- Brick Breaker
- Ultimate Tic Tac Toe
- Game of Life
- Transport Management System
- Character Crush

Honors and Awards:
- Deans List of Honor's, Fall 2020 & Spring 2021
- Rectors List of Honor's, Fall 2021, Spring 2022, Fall 2022 & Spring 2023

Interests:
- Web Development
- App Development
- MERN Stack

Languages:
- English (Fluent)
- Urdu (Native)
```

Figure 9.10: Structuring Resume Result 1

This is the structured result we received from LLM for a resume

```

FIGURE 9.11: Structured representation of the resume.

Contact Information:
- Name: Fatinah Salman
- Email: fatimahsalman776@gmail.com
- LinkedIn: https://www.linkedin.com/in/fatima-salman-1836b11b5/
- Phone No: 0335-4477366

About Me:
- Proactive and smart working learner
- Enjoys taking up new challenges and solving complex problems
- Open to exploring and grasping new skills

Education:
- BS Computer Science at FAST NUCES Lahore (Aug 2020-July 2024, expected)
- CGPA: 3.99
- Currently in 6th semester
- Studied core programming subjects like Fundamentals, OOP, Data Structures, Operating Systems, Algorithms, Database Systems Software Design and Analysis
- Intermediate from Punjab College (Sep 2018-June 2020)
- Matriculation from Lahore Grammar School (Jan 2016-July 2018)

Skills:
- Programming: C++, Python, JavaScript (basics), SQL, C
- Web/App: React, HTML, CSS
- Tools: Visual Studio, Atom, MS Word, PowerPoint, Google

Additional Courses:
- The Complete REACT with Redux by Stephan Grider, Udemy (July 2022 - Sep 2022)
- AWS Course of Cloud Computing

Experience:
- Teaching Assistant for Spring 2023 Operating Systems under Mr. Saifullah Tanvir
- Responsibilities included leading discussions and review sessions, grading assignments and quizzes, providing individual support to students, and assisting with course planning

Projects:
1. Basic Banking System
2. Employee Management System - SQL and C#
3. Portfolio Website
4. A complete functional terminal in C

Achievements:
- Rector's list of Honors
- Dean's list of Honor (Fall 2020, Fall 2021, Spring 2022, Fall 2022, Spring 2021)
- SGPA: 3.96

Extra Curricular Activities:
- Manager at Career Counseling Society
- Exceptional Volunteer for SOFTEC programming Competition 2022
- Host for Official events of the university like Convocation and Prize Distribution Ceremonies.

```

Figure 9.11: Structuring Resume Result 2

This is another structured result we received from LLM for a resume

9.5 Achieving Consistency in Structuring Resumes

This experiment was to address the challenges identified in previous experiments related to inconsistencies in parsing resumes. We introduced predefined criteria classes. These classes were designed to cover all possible categories in a resume, such as education, experience, skills, etc. Each class was explicitly defined in the prompt. This explanation was to ensure that LLM correctly categorized the resume content into these predefined classes. We used Pydantic Parser from Lang Chain to enforce the generation of these classes. LLMs response was improved to a great extent. The success of this experiment suggests that structured approaches, combined with clear definitions can significantly enhance the LLM's performance in tasks requiring detailed categorization.

```

resume_prompt = """
Interpret and structure the content from text={resume}. If any details are not present in the text, do not make assumptions

1. Contact Information:
Extract applicant's contact details under the following headings:
- Name
- Email
- Phone Number
- Address
- LinkedIn
- Github

2. Education:
The text may contain educational milestones such as School, College, Undergraduate/Bachelors, Postgraduate/Masters, PhD, etc.
For each educational entry, classify under:
- Degree
- Institute
- Graduating Year
- Grade
- Details ( Any extra information provided like courses)
Entries should be sorted from the most recent to the earliest. If education is ongoing, specify this in the "Graduating Year"

3. Work Experience:
Identify work experiences, excluding those related to academic or university societies, events, clubs, or volunteer roles.
For each experience, classify under:
- Domain (e.g., marketing, HR, teaching, web development, AI)
- Position
- Organization
- Duration (Compute by subtracting start date from end date. If end date is missing, assume it's ongoing. If both start and end dates are provided, compute the duration)
- Details (Any extra information provided)
Again, sort experiences from most recent to earliest.

4. Extra-curricular Activities:
Identify roles related to academic or university societies, events, clubs, or volunteering.
Classify each activity under:
- Domain (e.g., volunteer work, competition, society membership)
- Position
- Organization
- Duration (Compute by subtracting start date from end date. If end date is missing, assume it's ongoing. If both start and end dates are provided, compute the duration)
- Details (Any extra information provided)
Again, sort experiences from most recent to earliest.

5. Projects:
For each project the text presents, classify under:
- Name
- Domain
- Technologies Used
- URL

6. Skills:
Highlight prominent skills, whether explicitly mentioned or inferred from experiences and projects.
For each skill, mention:
- Name

7. Certifications:
Extract details on certifications or specialized courses (e.g., a React course from Udemy). Classify each certification under:
- Name
- Year
- Organization
- Domain (e.g., web development, machine learning)
- Details

8. Honors and Awards:
Identify accolades earned during the applicant's academic or professional journey. For each, classify under:
- Name
- Year
- Organization
- Domain (e.g., web development, machine learning)
- Details

9. Languages:
Determine languages the applicant is familiar with and possibly their proficiency level.
Classify each language under:
- Language
- Proficiency Level (e.g., beginner, intermediate, advanced)

{resume_format}
"""

```

Figure 9.12: Structuring Resume with Classes Prompt

This is the prompt we used to structure the resume into fixed categories

```

{
  "contact_information": {
    "name": "Maham Basit",
    "email": "mahambasit01@gmail.com",
    "phone_number": "03354388894",
    "address": "Lahore, Pakistan",
    "linkedin": "linkedin.com/in/maham-basit",
    "github": null
  },
  "education": [
    {
      "degree": "BS Computer Science",
      "institute": "FAST NUCES Lahore Campus",
      "graduation_year": "Present",
      "grade": "3.99 CGPA",
      "detail": null
    },
    {
      "degree": "GCSE Alevels",
      "institute": "Beaconhouse College Program Johar Town",
      "graduation_year": "2020",
      "grade": "4A*",
      "detail": null
    }
  ],
  "work_experience": [
    {
      "domain": "Software Engineering",
      "position": "Intern",
      "organization": "Conrad Labs",
      "duration": "2023",
      "detail": null
    }
  ],
  "extra_curricular_activities": [
    {
      "domain": "Competition",
      "position": "Data Structures and Data Bases Competition Coordinator",
      "organization": "SOFTEC",
      "duration": "2023 - Present",
      "detail": "Biggest IT event held at FAST NU Lahore"
    },
    {
      "domain": "Volunteer",
      "position": "Team Lead",
      "organization": "Future Fest",
      "duration": "2023",
      "detail": "Biggest IT conference event in Pakistan"
    },
    {
      "domain": "Society",
      "position": "Assistant Vice President",
      "organization": "STACKS FAST Chapter",
      "duration": "2022 - 2023",
      "detail": null
    }
  ],
  "projects": [
    {
      "name": "Textile Mill Salary Management System",
      "domain": "Database",
      "technologies": ["SQL", "C#", "HTML", "CSS", "Asp.net"],
      "url": null,
      "detail": "Project with database concepts of SQL, backend in C# and frontend in HTML CSS on Asp.net framework."
    },
    {
      "name": "Brick Breaker",
      "domain": "Game Development",
      "technologies": ["C++"],
      "url": null,
      "detail": "Console based game in C++ designed using concepts of Programming fundamentals and basic graphics."
    },
    {
      "name": "Ultimate Tic Tac Toe",
      "domain": "Game Development",
      "technologies": ["C++"],
      "url": null,
      "detail": "Console based game in C++ designed using concepts of Object Oriented Programming."
    }
  ],
}

```

Figure 9.13: Structuring Resume with Classes Result 1

This is the structured result we received from LLM for a resume

```

{
  "contact_information": {
    "name": "Fatimah Salman",
    "email": "fatimahsalman776@gmail.com",
    "phone_number": "0335-4477366",
    "address": "Lahore, Punjab",
    "linkedin": "https://www.linkedin.com/in/fatima-salman-1836b11b5/",
    "github": null
  },
  "education": [
    {
      "degree": "BS Computer Science",
      "institute": "FAST NUCES Lahore",
      "graduation_year": "July 2024(expected)",
      "grade": "CGPA: 3.99",
      "detail": "Currently in my 6th semester. Have studied core programming subjects like Fundamentals, OOP, Data Structures"
    },
    {
      "degree": "Intermediate",
      "institute": "Punjab College",
      "graduation_year": "June 2020",
      "grade": "Marks: 991/1100",
      "detail": null
    },
    {
      "degree": "Matriculation",
      "institute": "Lahore Grammar School",
      "graduation_year": "July 2018",
      "grade": "Marks: 1056/1000",
      "detail": null
    }
  ],
  "work_experience": [
    {
      "domain": "Teaching",
      "position": "Teaching Assistant",
      "organization": "FAST NUCES",
      "duration": "Spring 2023",
      "detail": "Operating Systems- Mr. Saifullah Tanvir\n\nled discussions and review sessions, graded assignments and quizzes"
    }
  ],
  "extra_curricular_activities": [
    {
      "domain": "Academic Society",
      "position": "Manager",
      "organization": "Career Counseling Society",
      "duration": null,
      "detail": "Organized events like Annual Job Fair, NUACES Talks, Seminars, Webinar, recruitment drives etc"
    },
    {
      "domain": "Volunteering",
      "position": "Exceptional Volunteer",
      "organization": "SOFTEC",
      "duration": "2022",
      "detail": "Exceptional Volunteer for SOFTEC programming Competition 2022"
    },
    {
      "domain": "University Events",
      "position": "Host",
      "organization": "Official events of university",
      "duration": null,
      "detail": "Hosted official events of the university like Convocation and Prize Distribution Ceremonies"
    }
  ],
  "projects": [
    {
      "name": "Basic Banking System",
      "domain": null,
      "technologies": null,
      "url": null,
      "detail": null
    }
  ]
}

```

Figure 9.14: Structuring Resume with Classes Result 2

This is another structured result we received from LLM for a resume

9.6 Conclusion

This section mentions multiple experiments that we carried out to test the capabilities of Large Language Models. Experiments varied from simple recruitment decisions to complex real life scenarios. The result showed that use of advanced prompting techniques like role based, chain of thought prompting can help LLM to generate accurate results.

Chapter 10 Conclusions

Our web-based application Recruit Ease is being designed to optimize the recruitment process by harnessing the capabilities of LLMs and prompt engineering. The major task is to optimize the different stages of the recruitment process like resume screening and interview feedback. Our goal is to make the hiring process more efficient and smooth. In this phase, we have developed a prototype that demonstrates the potential of LLMs in enhancing recruitment tasks and successfully implemented some important functionalities of the application.

The prototype aligns well with our initial objectives. Features like user authentication, job description management, and resume processing, are included. However we have limited the the scope of our application to a simple web application without mobile responsiveness and we have excluded the integration with other platforms as its not a part of our scope. One of the major challenges faced was in refining the perfect prompt for LLMs that helps us in achieving the desired results as we wanted to ensure that the resume analysis is both accurate and efficient.

Looking forward, in FYP-2, our plan is to expand the application's features and implement all of the functional requirements, presenting an application that is ready to be deployed and used in any company. We also aim at conducting extensive testing for robustness, and other parameters focusing on improving the overall user experience.

In conclusion, we have formed the proof of concept needed before building the entire application. Recruit Ease will be using the techniques for prompt engineering and will be harnessing the capabilities of LLMs. The work done so far has help us in understanding the potential complexities that could be faced and the different strategies that could be deployed to solve them. The goals and objectives of the application are clear and the prototype helps in demonstrating the practical feasibility of using LLMs for the major processing in the recruitment process.

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